

Some documentation for
Minos (V2)
Contest Logging Software
(V2.0.8.0 Beta)



This document is based on,
and specifically relevant to,

V2.0.8.0 Beta

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The **Minos Contest Logger** is supplied under Version 3 of the GPL or any later version.

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The **Minos Contest Logger** is **not** a supported product; no commitment or liability is or will be accepted to address any problems that may be encountered in using it.

However, any bugs or deficiencies should be notified to assist in the continuous development and improvement. If you use Minos and want to discuss the program or have any problems with it you are encouraged to join the project discussion mailing list on Groups.io.

Either visit <https://minos.groups.io/g/main> or send a blank email to main+subscribe@minos.groups.io

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Preface

This document is intended to provide information ranging from simple installation and usage through to detailed internals, so don't worry if some of it is of no use/interest to you. There have been extensive changes since Minos V1. With V2 the software is (usually) installed (see section 1.1), and the Applications (see section 4) are a major new facility.

As of V2.0.7 the user has the ability to configure the layout of the Logger main panel (see section 3.5). Unless otherwise stated, all descriptions and illustrations in this document use the default configuration.

This is a 'live' document that **may** be revised on the basis of any comments, feedback, or additional information received (via the project discussion mailing list). No commitment or liability is or will be accepted to address any problems that may be encountered in using it.

The software it describes is also under continuing development, so may not exactly match what is described here.

This is the latest release of the document for software that is currently under development. As with software, expect bugs; some of these will be known, others will not. **Please report any errors that you find (after checking that you have the latest version of the document).**

There is an implicit assumption that readers and users of the software are 'PC literate' and know about windows, panels, scroll bars etc.

The Logger runs on Windows 7 or later, but the document is written from the perspective of running the programme on Windows 10 Home (various versions, latterly 1803). At this time there are no known problems with running it on other versions, indeed it has been extensively tested on a number of other platforms, but no guarantees are given.

The Logger also builds and runs native on Linux. The intention is to also release binaries for Ubuntu (32 and 64 bit), and Raspberry Pi. Further platforms (other varieties of Linux, Android, Mac) may *eventually* be added. It is yet to be decided what will happen with documentation for these other platforms.

The general usage should be the same, or very similar; any significant differences will be noted as and when discovered/reported to the author.

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Minos V2.0.8 Beta

1 Introduction

The Minos Contest Logger is a logging program originally written by Mike Goodey G0GJV, now being developed by G0GJV and others. All RSGB and Region 1 VHF and UHF amateur radio contests are supported. The Logger supports real time or post event data entry and production of an entry log file for submission to the adjudicators.

Operation of the program is intended to reflect the normal practice of filling in a log book. The fields appear in that order on the screen, and are below a similarly laid out list of previous contacts - similar to a log book.

It is mainly intended for **on-line** log entry, but can be used to enter logs post event. The less done after the event the better; all relevant data should be captured during the contest.

A single operator can't keep both computer and paper up to date together. With only one operator, keep the computer log up to date - it is your duplicate check system as well as the log that will eventually be used to submit your contest entry.

Where possible a second operator/logger should use the computer, and the main operator should maintain a traditional paper log. This way, the main operator always has a piece of paper to scribble notes on, as well as a providing a backup log.

The user is ALWAYS right - even if insisting that a W9W/MM worked him from ZZ99ZZ square. Although errors may be reported, in the end the program can be made to log it like that anyway. However, a report of 6 and 0, or invalid dates and times will **not** be allowed.

Error messages should never get in the way of entering more data. So, except in unusual circumstances, the program will not display error messages that must be acknowledged.

For more on the history of Minos, and its predecessor GJVLog, see <http://minos.sourceforge.net/history.html>

1.1 Installation

The Minos installation programme (**MinosInstall_2_0_8_Beta.exe**) is downloaded (and saved to some convenient location) from <https://sourceforge.net/projects/minos/files/>

It is recommended that Minos V2 is installed to a different location than that used for Minos V1, even after release. Since V2.0 there have been, and continue to be, significant changes in the way configuration is handled. It is prudent to save a copy of the \Configuration folder prior to installing the new version in case you want to revert back.

Updates to the Beta can be installed to the same location as an existing V2 or later Beta. **However**, be aware that the format of the Rig and Rotator configuration files has changed during development so earlier versions should be deleted and the details set up again. If you fail to do this an error message will inform you of the problem when you invoke the application(s).

Administrator privileges are needed to invoke the installation programme. The usual ‘changes to system’ dialogue may be seen, depending on system version and configuration:

- Open File – Security Warning (click Run)
- User Account Control ‘changes to device’ (click Yes)

The standard installation dialogue then follows, as shown below.

Accept the licence agreement:

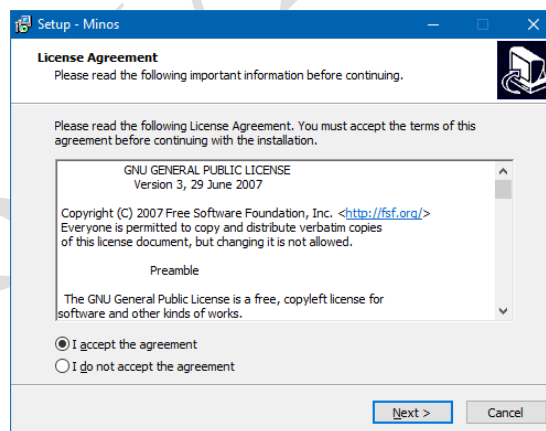


Figure 1 – Licence agreement

The default location for the installation is C:\Minos. This can be changed, but if this is done via the ‘Browse’ button be sure that it ends up where you want (by default a sub-folder is created).

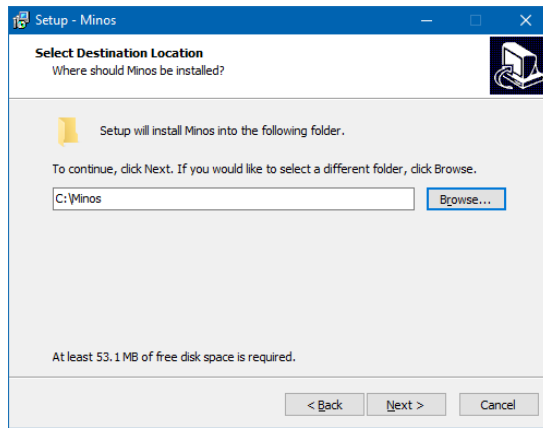


Figure 2 – Target location

You may of course get a message that the folder already exists, or needs to be created.

The log (.minos) files are by default placed in a sub-folder to the programme ‘root’ directory (see a description of the folders in Section 5.1).

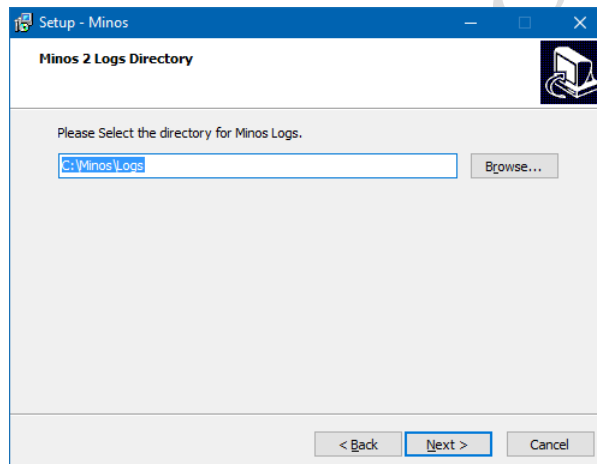


Figure 3 – Location of logs

By default a start menu entry is created, but this can be suppressed if so desired.

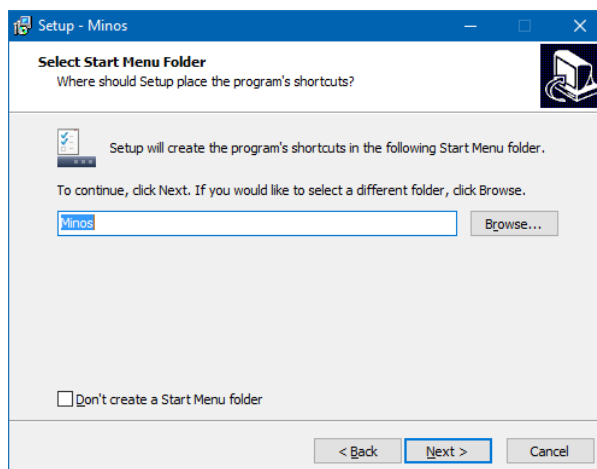


Figure 4 – Start menu

The “Additional Tasks” panel (Figure 5) gives the option of creating a desktop shortcut for invoking the Logger in a convenient fashion.

There is also an option to associate the “.minos” extension with the Logger so that double-clicking a log file will open the Logger.

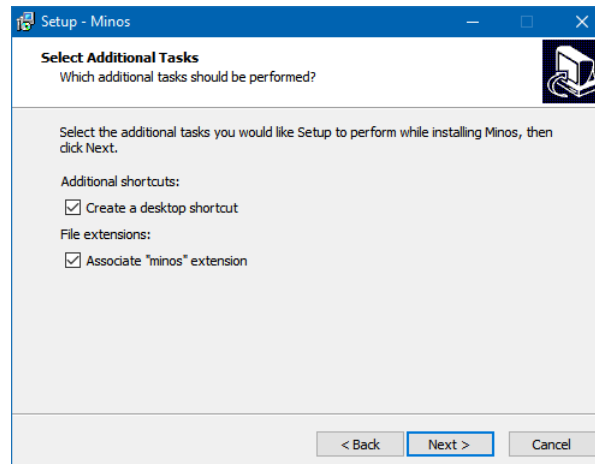


Figure 5 – Additional Tasks

Then there is a final check before installation starts:

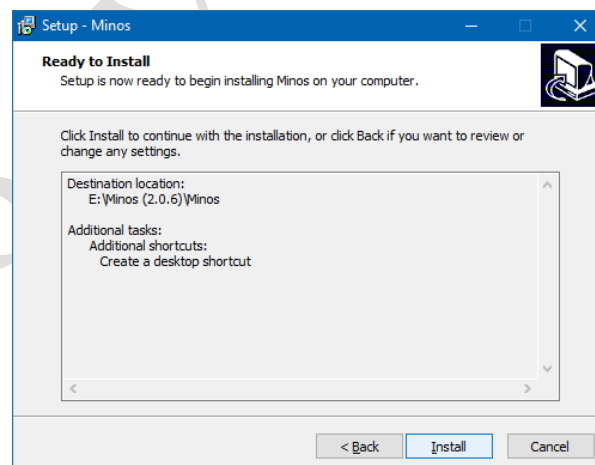


Figure 6 – Ready to install

After a short period of activity the completion notification (Figure 7) gives you the option to run the programme immediately. If this is not your first installation you may want to copy over configuration files before starting the programme.

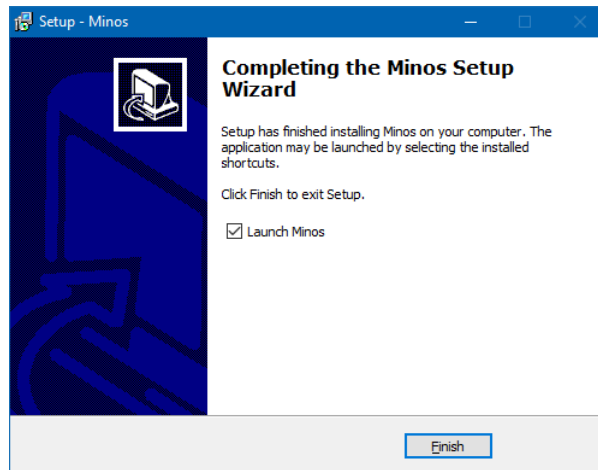


Figure 7 – Installation complete

1.1.1 Uninstallation

The Logger is uninstalled using the standard “Control Panel/Programs and Features” mechanism.

This removes all the binaries (\Bin), and the general Configuration information (that is not specific to the particular station installation), leaving (under the root folder):

- \Configuration: Station specific (The Display, Entry, ListPreload, LogsPreload, MinosConfig, MinosLogger, MinosRotatorConfig, QTH, and Station INI files
- All application configuration such as defined rigs and rotators is also saved, including screen layouts in ScreenConfigs.json.
- \Lists: (CSL files).
- \Logs: (Minos files).
- \TraceLog: (diagnostics).
- Changes to fonts and panel layout for the Logger will be retained in the Registry¹.
- Any personal files placed in the \Bin folder will be retained (and hence the folder).
- Any other folders that the user has created under the main Minos folder

1.2 Basic Configuration

Ensure that the computer clock is reasonably accurate (within 1 or 2 seconds or better, particularly for MGM contests), and set to BST if applicable (as can be seen at the bottom right of Figure 13, Minos calculates and displays UTC² - in the Auxiliary panel as well).

In order to perform the initial set-up, invoke the program and the Splash Screen will be displayed.

¹ These can be reset using “File/Exit MinosQT and Clear registry”.

² See section 3.4.4 for information about time adjustments that can be made.

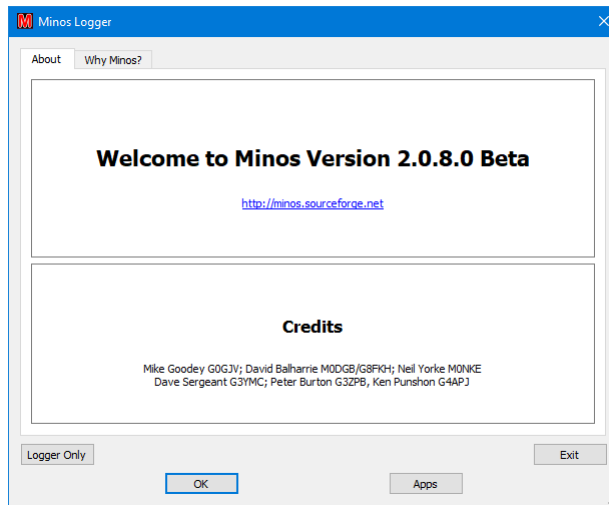


Figure 8 – Welcome screen

Click on OK³ (or “Logger Only”) to dismiss the splash screen. **Ignore the Apps button for now.**

You will certainly want to expand the Logger main panel. Depending on your screen size and resolution you may want to expand to full screen, particularly if you have a wide screen.

Click on “File/New Contest” (see Figure 9) to be presented with “Details of Contest Entry” screen (see Figure 10):

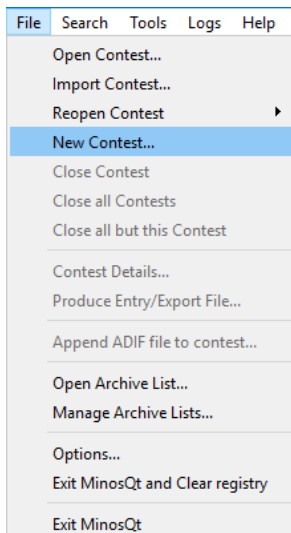


Figure 9 – File options

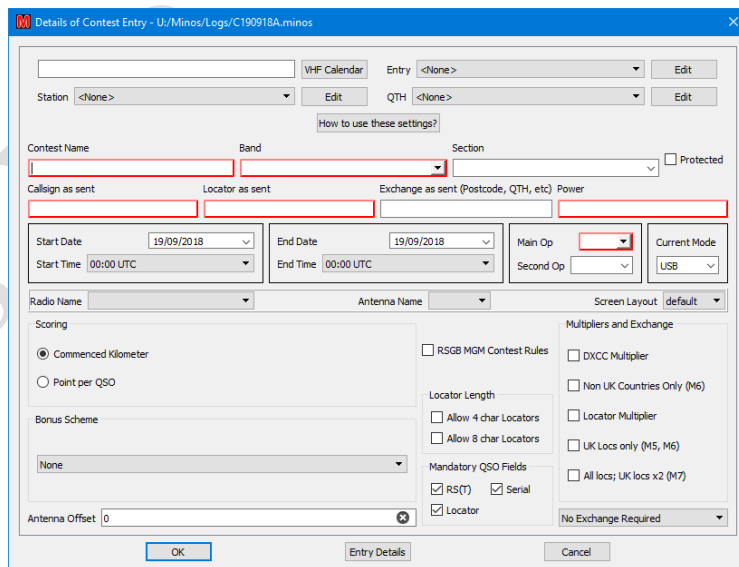


Figure 10 – Contest Entry Details

This then allows the set-up of various items that will be used in future. The red outlines indicate the boxes that still require information before a contest can be validly created. When you are creating a contest the red edgings will disappear as the required information is supplied.

³ You may be interested to read “Why Minos” on the “About” panel for background on the origin of the name.

Click on VHF Calendar and then (because it can, and does, change) “Download Latest Calendar”. At the time of writing, Minos ships with **calendar version 18.013 29/05/2018 for the remainder of this year**. Please check for updates before use, and certainly in January each subsequent year.

Clearly a new download of the calendar will be needed on **at least** a yearly basis⁴.

For now, close the VHF calendar in order to edit the grouped settings⁵, which can be applied to a contest all in one go. There are four basic groups of settings:

- **Entry:** All the extra bits for a real entry (entrant callsign, group name, contact details) which are stored in Entry.ini
- **Station:** Details of the rig, power, and antenna (stored in Station.ini); you will typically require one of these for each band, if only due to antennas.
- **QTH:** Where the station is located for this contest, including Maidenhead Locator used. The data is stored in QTH.ini.
- **Contest:** Typically derived from the contest calendar, specifying contest name, band(s), and scoring. If the calendar is not used then individual boxes can be configured (see Figure 12) or modified from the calendar derived values.

To define these settings, click on Edit, then select New Setting⁶. Assign a meaningful name and fill in the requisite data. A typical collection of grouped settings might for example be called:

- **Entry:** Self, Club, Contest Group
- **Station:** 50MHz, 70MHz, 144MHz, 432MHz, Portable-144MHz
- **QTH:** Home, VHF NFD, UKAC Portable

Click on the setting name on the left to select an existing setting and then its components are shown in the right hand pane, and can be edited individually. There is **no** validation of field content at this time, although errors may be highlighted later.

⁴ The intention is that this facility should be available on an indefinite basis, but there is no commitment to such activity.

⁵ If you have created these entries for an earlier version of the program you can copy over the INI files (to the Configuration folder). It is also useful to copy over any CSL files you may already have (to the Lists folder).

⁶ Once you have created entries, an existing setting can be copied and minor changes made to cope with such things as the same rig being used on multiple bands.

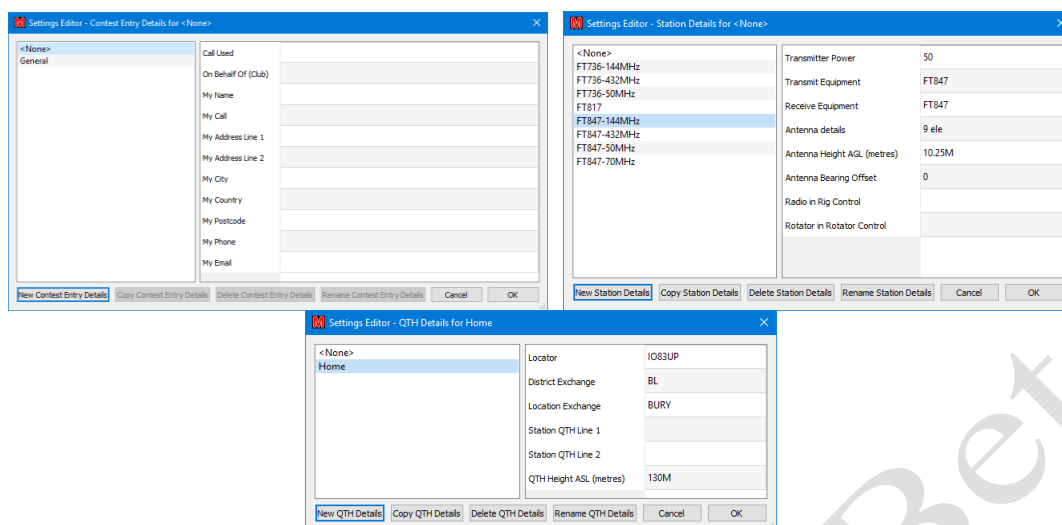


Figure 11 – Example Grouped Settings

Once the settings have been defined, use “Entry Details” to check that meaningful and complete data has been supplied.

During a contest “File/ Contest Details” can be used to check, and potentially change, these details (e.g. to add operators⁷ or comments).

For now, “Cancel” the contest to finish set up; this will lose any other data entered into the contest (but you will do this well before the contest won’t you?).

⁷ Operators can also be added via the pull down lists in the data entry panel.

2 Normal operation

Invoke the program from the shortcut on your desktop or the start menu⁸.

Any contest files that were left open last time will remain open⁹ (see section 3.4.1 for discussion of multi-band contests that would typically use this facility). Similarly, any Archive (CSL) files (see section 3.4.3) will be re-opened; these are implicitly Read-Only unless you deliberately edit them.

For normal single contest operation close anything other than the current contest, or ensure that the non-current contests are marked as “Protected” to prevent you accidentally modifying them. This is done by selecting “File/Contest Details” for the requisite contest and then ticking the “Protected” tick box on the right hand side just above the “Power” box, see Figure 12.

The protected status is preserved between invocations, and hence must be (temporarily) removed if you wish to change the content. This is done by clearing the “Protected” box, but note that it will revert to protected on the next invocation of the programme.

If a contest has been protected then all the “Contest Details” except for the “Protected” field are greyed out and any connection to RigControl or Rotator will be disabled.

If want to re-open a contest that you recently closed then “File/Reopen Contest” gives a list of the last 5 contests opened¹⁰. Opening one of the files on the list causes the entry to be removed, but it will be placed back at the head of the list when the log is closed again.

There are no known limitations (e.g. number of contacts, bands, field lengths....), but if you discover any please notify the support group (see Preface). Although it is possible to open at least 20 concurrent contests the ‘tab space’ at the top of the screen (on a wide, 1920x1080, screen) is full when 10 are open and it becomes complex to switch between them¹¹.

2.1 Creating a new contest

Use “File/New Contest” which will bring up the Contest Details panel as described in section 1.2.

Select the contest from VHF Calendar (normally for data entry during the contest; see section 2.2.1 for details of post-contest data entry).

Once Applications have been defined (see section 4.1) the “Radio name” and “Rotator name” pull-down lists can be used to define the default radio/rotator to be used for the contest. The connection can also be

⁸ The logger can also be invoked by directly clicking on the image (MqtLogger.exe in the \Bin folder) or by double clicking a log (.minos) file if the file association has been set up during installation.

⁹ A warning message will be given if the file cannot be found.

¹⁰ If the file that you attempt to re-open has been deleted then no error message will be given, but the entry will be removed from the list.

¹¹ From the operator perspective 5 is probably more than enough!

established/changed through the QSO entry panel (see Figure 61 and Figure 79) – at which point these boxes will be filled.

Select the requisite bundled settings from the drop down lists for Entry, Station, and QTH. Any bundles set to “<None>” will be ignored (and you will have to specifically enter the data before submitting the entry).

If the setting you want isn't there, use the “Edit” button to define a new setting or copy and then modify an existing setting in the manner described in section 1.2.

Note that the “Section” selected is no longer relevant to RSGB contests; it is determined by the Contest Entry Upload Robot. However, you may want to enter the requisite value here for completeness of the log¹² or for entering a non-RSGB contest.

If the contest is not selected from the Calendar then use the relevant fields on the “Details of Contest Entry” panel (see Figure 12) to set the name, date/time, scoring, multipliers etc. Any bonus scheme in use must also be selected from the pulldown list at the bottom left.

The figure shows two overlapping windows from a software application. The larger window in the background is titled 'Contest details' and contains several sections. The 'Scoring' section has 'Commenced Kilometer' selected. The 'Bonus Scheme' is set to 'None'. The 'Antenna Offset' is 0. The 'Locator Length' section has 'Allow 4 char Locators' and 'Allow 8 char Locators' checked. The 'Mandatory QSO Fields' section has 'RS(T)', 'Serial', and 'Locator' checked. The 'Multipliers and Exchange' section has 'DXCC Multiplier', 'Non UK Countries Only (M6)', 'Locator Multiplier', 'UK Locs only (M5, M6)', and 'All locs; UK locs x2 (M7)' checked. The 'No Exchange Required' option is selected in the 'Exchange Required' dropdown. A smaller window in the foreground shows the 'Bonus Scheme' dropdown menu with options: 'UKAC Bonuses (B4)', 'None', 'UKAC Bonuses (B2)', 'UKAC Bonuses (B4)', 'NAC Bonuses', and 'Antenna Offset'. The 'None' option is highlighted.

Figure 12 – Contest details and Bonus schemes

Use “Entry Details” to check that everything is complete and as expected. Any changes specific to the particular contest can be made via “Entry Details” at any point prior to exporting the log. If an error needs to be corrected it is possible to re-export the entry and re-submit (up until the closing date of the contest).

Click “OK” and the “Save new contest as” panel will generate a default name of the form **G4APJ_2017_09_02_144 MHz.minos** for the file located in the “Logs” folder¹³. In the event that the file already exists the file will have “_1” (etc.) appended. It is possible to specify some other name, or browse to another folder if you want, but this may introduce undue complexity if you are unsure of what you are doing.

If you have not provided sufficient information when you click “OK” the “save as” panel will not appear and the cursor will be moved to the first field identified as requiring input. All fields still required remain outlined in red.

Clicking “Save” to the “Save new contest as” panel causes the blank Data Entry panel to be displayed ready for entering details of QSOs (see Figure 13). The contest just created is highlighted (red text on tab)¹⁴.

¹² This can be done from the pull-down list defined for a contest from the calendar, or by entering any arbitrary text (which is completely ignored by the Logger).

¹³ Note that the Contest Name does not appear in the filename, regardless of how it was entered.

¹⁴ Protected contests show green and those not selected show blue.

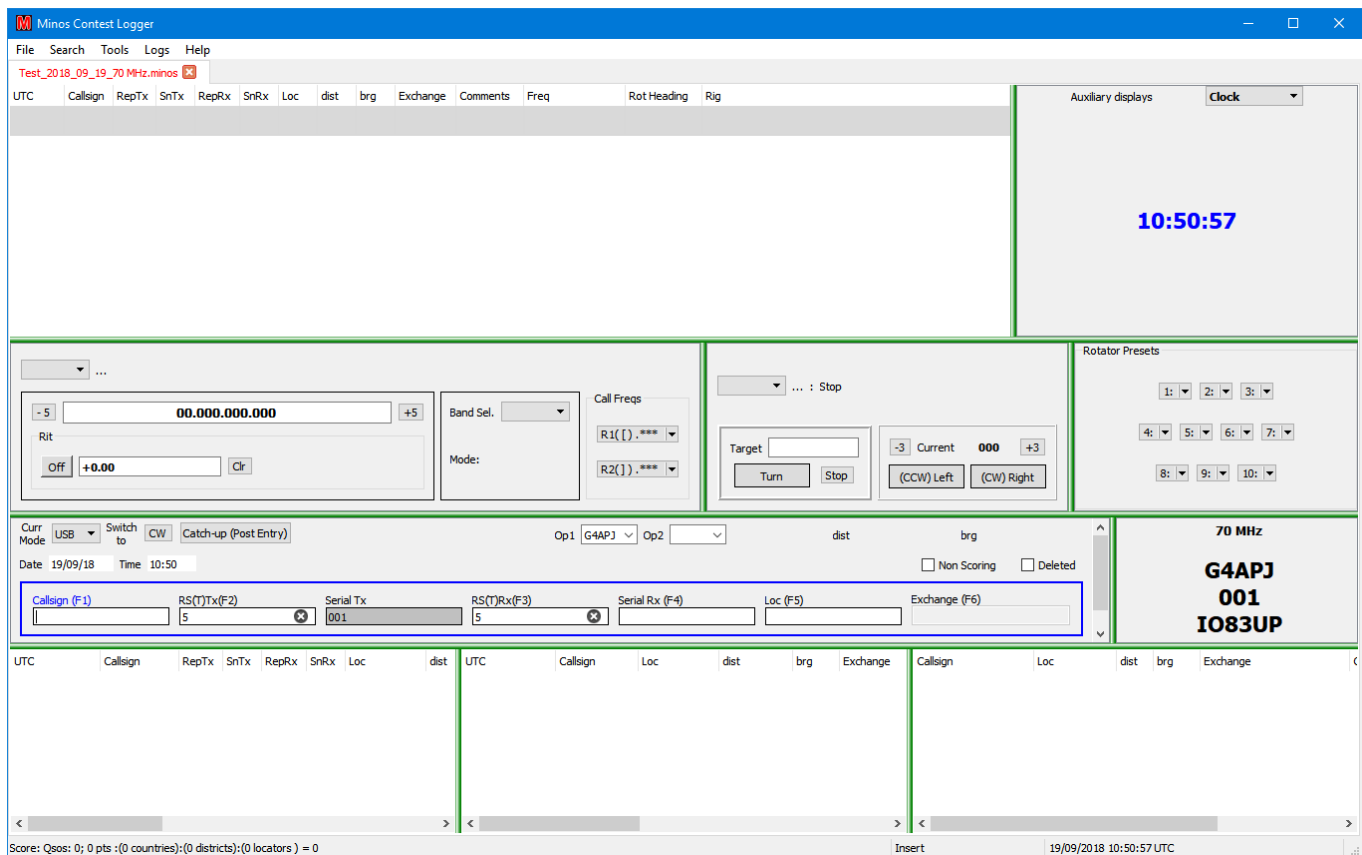


Figure 13 – Initial screen for entering QSO details

Figure 13 uses the default panel layout which includes the Rig and Rotator panels. See section 3.5 for information on how to change this default to remove these panels if they are not being used.

The width (and heights) of the sub-panels (with green edges) can be adjusted by sliding them across (up/down). Column widths can also be adjusted. The worked/un-worked filters and the fonts/panel locations/column widths are saved across invocations (and re-installation). Defaults can be reset using the “File/Exit MinosQt and Clear registry” option which resets all Minos components.

2.1.1 MGM Contests

In order to support the new MGM contests (beginning in April 2018) there are a variety of options to control QSO requirements (see Figure 14). These will be filled in automatically, and correctly, if the contest is selected from the VHF calendar (which will always be the preferred method).

Figure 14 – MGM Contest Details

Once the “RSGB MGM Contest Rules” box is ticked (either automatically from the selected contest or manually) it sets things up for the scoring of the RSGB MGM contests **and locks the options** to prevent changes.

This sets “Locator Multiplier”, “Allow 4 char Locators” (scores are calculated as centre to centre of the 4 character Locators), and clears Serial in the Mandatory Fields (i.e. serial numbers are not only not required, but cannot be entered¹⁵).

Once you set a QSO mode to MGM (in any contest) the validation on the report is removed - it just requires it to be present – **so be careful what you type**.

If the MGM software being used can create ADIF files (e.g. **WSJT-X ADIF Export**) the log may be imported into Mimos. Create the contest then select “File/Append ADIF file to contest” (see Figure 9) and browse for the .adi file to be included¹⁶.

Beware: the whole of the .adi file will be included so make sure the file only includes the data for the contest! Any records outside of the contest duration will be marked invalid but make cause problems with e.g. duplicate QSO marked within the contest.

If you have cause to use “Insert Before” or “Insert After” (Section 2.2.5) with an MGM contest don't worry about the SerialTx being blanked out, resulting in a serial of 000.

¹⁵ Any of the “Mandatory Fields” not ticked is prohibited.

¹⁶ Defaults to the Mimos log location.

2.2 Entering QSO details

When the Data Entry panel is displayed the cursor will initially be in the Callsign field; to return to that field click on it with the mouse or hit “F1”. By default the Mode (top left of data entry panel) is set to USB, but it can be changed by means of the associated pull-down list, or will track changes made to the physical rig.

There is no required sequence of data entry, although the default order of the boxes (as selected by the “TAB” or “RETURN” key) reflects normal contest exchanges.

“RETURN” is normally used to move between fields¹⁷, by selecting the next empty/invalid field¹⁸. For the RS(T) fields, if Autofill has been set¹⁹, the remainder of the field will be filled with 59 or 599 as appropriate to the mode (not for MGM); any data entered will not be overwritten.

Fields may be selected ‘out of order’ using the mouse or the designated Function key (F1/F2/F3/F4/F5/F6 in Figure 15); a field may be partially filled (although of course must be completed before the QSO can be normally logged – see section 2.2.2 for details of logging incomplete QSOs).

The screenshot displays the Minos V2.0.8 Data Entry panel. At the top, there are controls for the current mode (USB), a switch to CW, and a catch-up button. Below this, the date and time are set to 19/09/18 and 11:01. The main data entry area includes fields for Callsign (F1), RS(T)Tx (F2), Serial Tx, RS(T)Rx (F3), Serial Rx (F4), Loc (F5), and Exchange (F6). The current QSO being entered is G4APJ, 020, I083UP, at 144 MHz. Below the entry fields, there are three columns of contest data. The first column shows the current contest with 12 matches, including G0M1W/P, G8ZRE, G0BBB, G4ZTR, G0XDI, GW8IZR, GM4PPT, GM6JNJ, G8NEY, G8ONK, G3YDY, and G4HGI. The second column shows the [50 MHz] 50/70/144/432MHz Christmas Cumulatives Contest with matches like G8ZRE, G8ONK, G4HGI, G8ZRE, G8DTF, G4HDI, G16ATZ, G4HPI, G0XDI, GW8ASD, GM4PPT, G1SMI, G8NEY, G8ONK, G3YDY, and G4HGI. The third column shows the CSL-2018.csl contest with matches like G4FJC, G4ASR, G3NPI, G4CLA, G0EAK/P, G0FWU, G3PHO, G3TDH, G3WRA/P, G0VVE, G3XDY, G3ZPB, G4FEV, G3H, G3UVR, G4FOH, G0BFJ/P, G0ODQ, G0XDI, and G3WAG/P. The bottom status bar shows the score: 19 Qs, 19 pts, 4 countries, 0 districts, 10 locators, and 266. The time is 19/09/2018 11:01:37 UTC.

Figure 15 – Details of contact

Upper case is forced as data is entered in the Callsign, Loc, or Exchange²⁰ fields, and Minos displays possible matches from the current contest²¹, any other contests that are open, and any open CSL file(s).

¹⁷ Clicking on the “Log” button has the same effect.

¹⁸ For a new contact this will be the same sequence as if TAB were used.

¹⁹ This is most conveniently set under “Tools/Signal Report AutoFill” but can also be done through “File/Options” (see Figure 90).

²⁰ If Postcode multipliers or similar are in effect.

²¹ Duplicates are always displayed first.

On RS(T) fields in non MGM contests, only numeric characters and a tone of A are valid; you can enter e.g. 59A for auroral contacts. In MGM contests no validation of RS(T) fields is done. On serial number fields only numeric characters are allowed, except in the received serial number; “-” shows no serial number is received (e.g. for NAC contests) .

As a Locator is entered the distance and bearing is calculated and displayed in the top right of the panel. Partial Locators give approximate figures, designated by brackets.

Note that data is not validated as it is entered, only when you attempt to log the entry. This allows for partial information to be entered (e.g. during QSB) and then be enhanced/corrected later in the contact.

If a suitable match is highlighted (with the grey bar, see Figure 15) then the contents of the selected entry can be transferred to the data entry boxes by clicking on the “Match Xfer” button or hitting the “F12” key²² (or by double clicking the requisite entry). This *may* be faster than retyping the data, and should reduce typos (but beware, /P stations are not the only ones that may change location/locator details).

The content of individual fields may be deleted using the new “Clear” marker (black circle with white cross) at the end of relevant fields.

The current QSO may be abandoned, and all entry fields reset to default, by hitting the “ESC” key or clicking on the “Cancel” button. Hitting “ESC” or “Cancel” again recovers the fields, provided nothing has been typed since the original cancellation.

This allows the re-use of the current contact number; if the QSO has progressed far enough to make this undesirable tick the “Non-scoring” box before “Forcing” the entry to be logged.

There are a number of circumstances that might require “Non Scoring” to be used without “Force”:

- All the details have been entered, but you know that the other end missed them.
- To mark the FIRST contact with a station as a duplicate, rather than a subsequent one.
- The contact is known to be invalid – maybe with a station who will be one of your operators on the cover sheet.

Once all fields are filled with valid data, “RETURN” will log the contact²³. This happens when the program sees that there is valid data in each required field, which under some circumstances may be before you think the contact data is complete. In this situation use cursor up to select what is now the previous contact, and edit in the required additions.

If the QSO data is incomplete²⁴ and you cannot log normally, then click on “Force” which will result in the dialogue described in section 2.2.2.

²² Watch out: the callsign may also get amended from that entered (e.g. /P added).

²³ The ‘running totals’ at the bottom of the Logger will be updated (see Figure 13 and Figure 17).

²⁴ For instance, some stations may not know their locator; you can work it out later from the geographical location they give you and then edit the QSO details.

If your computer logging falls behind, and you miss one or more contacts, it is possible to override “Serial TX” by double clicking the field and entering the required next serial number. Minos will then check that you intend to do this and that you want to fill in the contacts later.

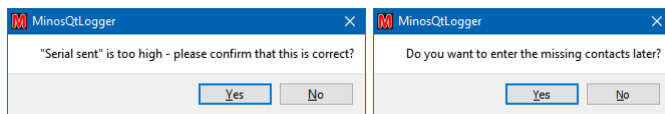


Figure 16 – Skipping contacts

Dummy entries will be created for the missing serial numbers. You can then fill in these missing contacts later (see section 2.2.4).

If you send some serial numbers twice, then double click and overtype the serial sent field to make the log reflect what you actually did.

2.2.1 Colour Coding

During normal operation various colours are displayed on the screen to indicate status or progress. Figure 17 shows an example part way through contest²⁵.

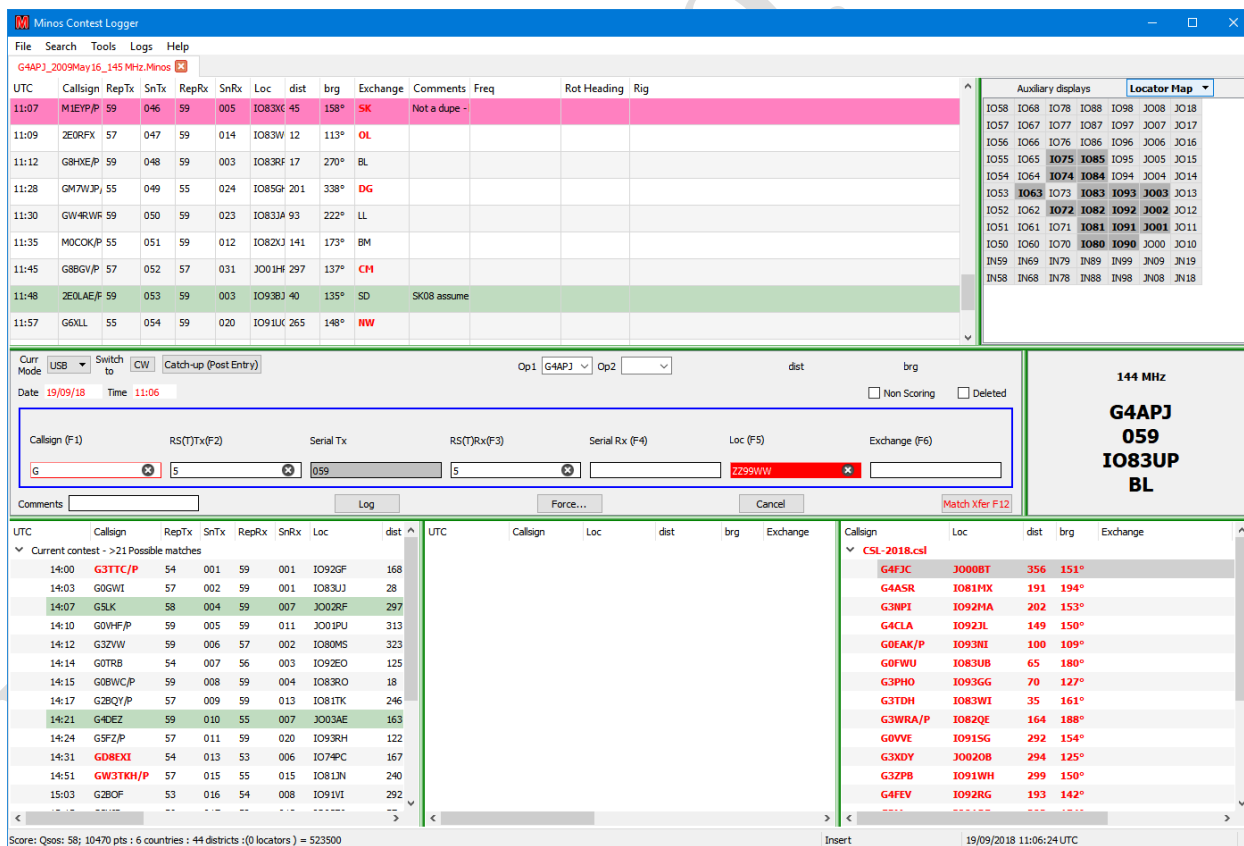


Figure 17 – Colour coding

²⁵ The screen layout has been modified from the default to remove the application panels. The image is also from a previous version that did not have the “clear field” facility.

On the main display panel:

- **Red Callsign:** New DXCC multiplier (where DXCC multiplier scoring active)
- **Red Loc:** New Maidenhead Locator (with Locator multipliers active)
- **Red Exchange:** New Postcode (or other Exchange, when multipliers active)
- **‘Greenish’ horizontal bar:** QSO where the details have been edited
- **‘Mauve’ horizontal bar:** record had to be forced; turns green once missing data entered
- **Colours in Locator stats:** Depending on the bonus scheme in use, various colours may be displayed²⁶; grey indicates no bonus scheme in use. Pale indicates unworked, bold indicates worked.

Within the data entry area:

- **Red Date or Time:** current value is invalid for the designated contest duration (outside the contest time limits).
- **Blue text** above the data entry boxes: indicates the currently selected field
- **Red box** around callsign or locator indicates incomplete data.
- **Solid red fill** shows possible invalid data (including duplicate callsign)

Inside a match box:

- **Blue horizontal bar:** Currently selected match that will be used in a transfer operation (using F12).
- **Red Callsign/Loc/Exchange:** As per main panel

The content of the bottom right panel (‘CSL match’) is in red to show that panel is selected. See Figure 15 for an example of another contest panel selected.

2.2.2 Forcing a contact

If the data for a QSO is incomplete and cannot be normally logged then the “Force” button (or “Alt/F”) can be used to store the current details pending further communication with the station or later calculation (e.g. when the other station does not know its Locator). This results in a panel as shown in Figure 18.

The current errors in the QSO are displayed, along with various options that may be selected:

- **Unfilled:** Data is missing
- **Backpacker:** If a back packer, or other station, has moved since you worked them earlier this allows the second contact to be marked as valid rather than a duplicate.
- **Insert manual score now:** (if you really want to).
- **Non Scoring:** (the default)
- **Don’t Print/Export Entry:** to logically remove a contact
- **Force Country for Multiplier:** To force the use of a particular country for country multipliers.
- **No District Available:** (‘auto’ for non GB stations)
- **Cross band:** half score

²⁶ Which also show up in the other panels displaying Locator information.

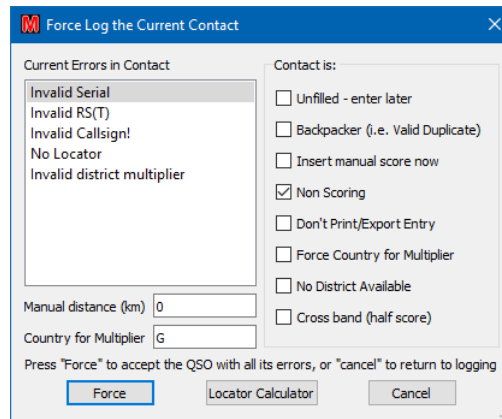


Figure 18 – Force log dialogue

The “Current Errors in Contact” may also indicate “Locator probably not within country”.

Clicking “Force” (check focused, so also RETURN) actually writes the record.

2.2.3 Modifying a previous contact

As and when more data is forthcoming for a Forced log entry the record may be edited to enhance or correct the QSO entry.

Double click the requisite entry, which results in a display of the form shown in Figure 19.

Figure 19 – Editing the details of a QSO

Once the data has been completed, ensure “Non Scoring” has been cleared (just above the Locator field) before logging the record by clicking “Log”²⁷. If the data is still not complete then “Force” will invoke the force panel dialogue.

²⁷ Note that if the serial number sent is changed it is assumed that it is a correction and no intermediate serial numbers will be offered to be created.

The modified entry in the main log display turns green (see section 2.2.1 for information on colour coding of records) if all information has now been supplied.

A message box (Figure 20) allows confirmation that the changes should be applied. For valid entries the distance and bearing are then calculated and the record is included in the score.

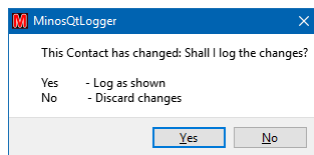


Figure 20 – Confirmation of changes

“Prior”, and “Next” may also be used to navigate between entries that still require amendment. If the current selected entry has been modified when one of these buttons is selected then the confirmation of changes dialogue will appear before the action is taken. If “No” is selected then the original record will be restored and remain selected.

“Return to Log” **saves** any changes made and finishes the QSO editing activity.

Use “X” to **discard** any changes that you have made and return to the log.

2.2.4 Completing unfilled contacts

If unfilled contacts have been created then the “First Unfilled QSO” button will appear within the QSO data entry panel (see Figure 21).

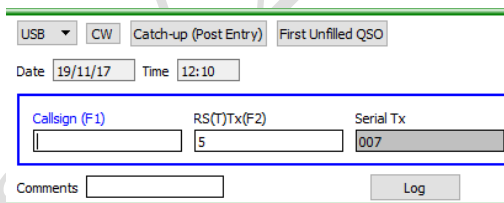


Figure 21 – First Unfilled QSO button

Click on this button to locate the unfilled QSOs. If any are found a similar dialogue to the edit QSO will appear (see Figure 23). Correct the problems, if only by marking the QSO as non-scoring, then use “Next” to move on until all have been processed.

There shouldn’t be any left in the log with “UNFILLED CONTACT” in the comments, nor any incomplete information marked in ‘mauve’!

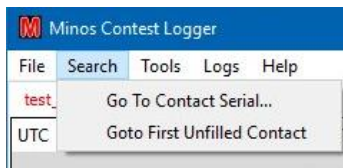


Figure 22 – Search pulldown menu

Figure 23 – Completing unfilled contacts

“Return to Log” **saves** any changes made and finishes the QSO editing activity.

Use “X” to **discard** any changes that you have made and return to the log.

2.2.5 Insert Before/After

When editing a QSO or filling any unfilled QSOs the “Insert Before” and “Insert after” buttons allow the creation of new QSO records in the indicated position relative to the currently selected QSO using the “Edit QSO” dialogue.

These buttons only become active after the first two records have been written, and do not allow additions before the first record or after the last.

2.3 Submitting an entry

Select “File, Produce Entry/Export File” and check that the ‘administrative’ details displayed are complete and correct (see Figure 24). Any last minute changes made at this point, e.g. overall comments, are written to the EDI file so that they get seen by the adjudicator; they do not necessarily appear on the claimed score page.

Date Range (Calculated)	11/04/17;11/04/17
Contest Name	432MHz UKAC
Band	432 MHz
Entrant name (or group)	Ken Punshon
Station QTH 1	
Station QTH 2	
Section	AR
Callsign	G4API
Locator	IO83UP
Exchange/code/QTH sent	
Transmitter	FT847
Transmit Power	50
Receiver	FT847
Antenna	19ele
Height above ground	11.5M
Height above sea level	130M
(From QSOs) Operators Line 1	G4API
(From QSOs) Operators Line 2	
(Entry)Operators Line 1	
(Entry)Operators Line 2	
Conditions/Comments	
Conditions/Comments	
Conditions/Comments	
Conditions/Comments	
Name for Correspondence	Ken Punshon
Callsign for Correspondence	G4API
Address 1 for Correspondence	
Address 2 for Correspondence	
City for Correspondence	
Country for Correspondence	England
Postcode for Correspondence	
Phone number for queries	
email address for queries	G4API@Kesuue.org

File Format
☒ RegITest (Entry)
☐ ADIF (Export)
☐ G0GJV (Export)
☐ Minos (Export)
☐ KML (Google Earth)
☐ Printable (Export)

☐ Don't export serials (NAC, RegITest)

OK
 Cancel

Figure 24 – Checking the Entry data

Select the required format (for contest submission this is unlikely to be anything other than Reg1Test that is accepted by the VHFCC up-loader).

The other formats that may be used are:

- **ADIF:** to transfer data to other software packages
- **G0GJV:** (probably now redundant) to transfer to the old DOS Logger
- **Minos:** produces a ‘cleaned up’ log with all edits log cleaned out. It may also be used to produce a shorted abstract, e.g. for a Trophy Contest from a full IARU 24 hour event.
- **KML(GoogleEarth):** If you have GoogleEarth installed, generate one, and double click on it. It gives you all the contacts as pins on GoogleEarth.
- **Printable:** produces a plain text file for checking or printing

As can be seen in Figure 24 there is an option to suppress the export of serial numbers (NAC, Reg1Test).

Click “OK”, presented with “Save as” dialogue.

Select requisite folder and filename (.EDI). You may want to have your own naming scheme if intending to save these files. The default location for the EDI file is the same as the log files.

Click “Save” to write the file.

Check the content of the resultant file. Note that in an EDI file any per-QSO comments will have been abstracted and collected under the [Remarks] header in order to comply with the EDI field definitions; this can be confusing at first sight.

Go to the entry robot (<http://www.rsgbcc.org/cgi-bin/vhfenter.pl>) to submit your entry (you will get a copy back from the robot by email). If you wish to practise the submission process then take a look at the new VHF Log Test Facility on:

<http://www.rsgbcc.org/hf/rules/2018/rvhflogtest.shtml> and
<http://www.rsgbcc.org/cgi-bin/vhfenter.pl?Contest=Log%20Upload%20Tester>

To exit the program, use “File/Exit Minos” or “X”. If a contest is truly completed then it is advisable to close it first²⁸, possibly after marking it as Protected.

²⁸ The abstracted content of a contest may then be added to the archive list using one of the CSL utilities, or manually.

3 Additional Features

3.1 Post-event data entry

This facility is intended to allow transfer of data from a paper log. Do not use it to tidy up after the end of a contest.

Create the contest in normal fashion then click on “Catch-up (Post Entry)” in the Details of Contact panel (see Figure 25). The date will change to the start date of the contest (watch out in 24 hour, or longer, contest) and the cursor will move to the Time field which is set to 00:00 initially.

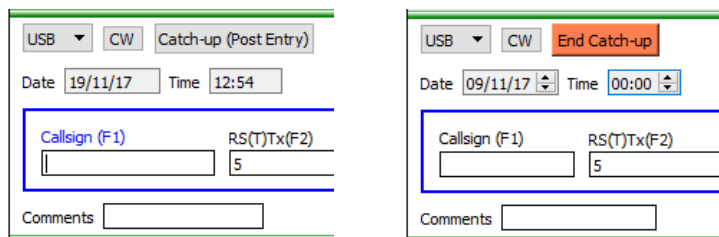


Figure 25 – Catch-up (Post Entry)

Enter the time (and date if needed) of the first contact and then enter QSO data in the normal fashion. The duplicate checking and CSL panels will operate in normal manner.

After each QSO is entered the cursor is set on the last character of the time field to allow amendment of the time to that of the contact. When finished click “End catchup” to return to normal operation.

Once all data entry is complete, submit the entry in normal fashion (see section 2.3).

3.2 Auxiliary displays

By default the top right of the Logger display is used to show auxiliary data in a single panel. “Tools/Screen Layouts” is used to control the number and position of such Auxiliary Displays (see section 3.5).

A pull down list at the top of each panel controls what content is displayed (Clock is the default). The panels selected are held within the contest context and are preserved when a contest is closed.

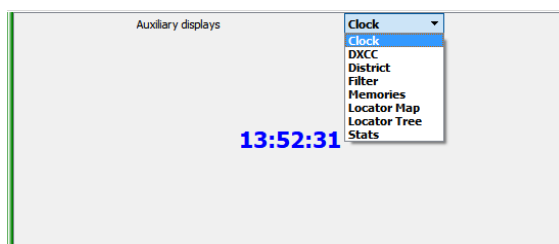


Figure 26 – Auxiliary display options

Different options may be selected to be displayed for each open contest, although any filters configured apply to all (relevant) contests. The option selected, and the filters applied, are tied to the contest and stored in the log (minos) file.

3.2.1 Clock.

The default is to display the current time in UTC. Blue means (the currently selected) contest is in progress, red means it has either finished or not yet started.

3.2.2 Statistics

- DXCC - displays the DXCC entities, under control of the selected filters. Most of the time only the European entities will be meaningful.

Auxiliary displays					
DXCC					
Call	Wkd	Locator	brg	Country	Other calls
EI	1	IO53XD	264°	Ireland	EI EJ
G	48	IO92GS	150°	England	2E G M
GD	1	IO74RE	293°	Isle of Man	2D GD GT MD ...
GI	1	IO64PR	295°	Northern Ireland	2I GI GN MI MN
GM	3	IO76VT	342°	Scotland	2A 2M GA GM ...
GW	4	IO82DG	212°	Wales	2W GC GW MC...

Figure 27 – DXCC Statistics

- District – In Postcode exchange contests can be used to display the Districts, under control of the selected filters.

Auxiliary displays					
District					
Code	Wkd	Locator	brg	District	
AL(G)	1	IO91TR	148°	St Albans	
BA(G)	1	IO81UF	180°	Bath(Frome)	
BD(G)	1	IO83XW	27°	Bradford(Skipto...	
BL(G)	2	IO83UO	180°	Bolton(Bury)	
BM(G)	2	IO92BL	168°	Birmingham	
BN(G)	1	IO90WT	154°	Brighton	

Figure 28 – District Statistics

- Filter – controls the display of the DXCC, and Districts content **only**. Selection of the geographical regions is meaningless for the Districts display, only worked/unworked are applied. These filtering controls get saved in the minos file for each contest. The panel select works per-contest

Auxiliary displays		Filter
Filters only apply to DXCC and District Continents only apply to DXCC		
☐ Africa	☒ Worked	
☐ Asia	☐ Unworked	
☒ Europe		
☐ N America		
☐ Oceania		
☐ S America		

Figure 29 – Statistics Filter

- Locator – displays Locators using a colour coding scheme to indicate the relative bonuses²⁹ (if bonuses are in use). The initial display of squares will be extended, and ‘sliders’ added, if a square outside of this range is worked.

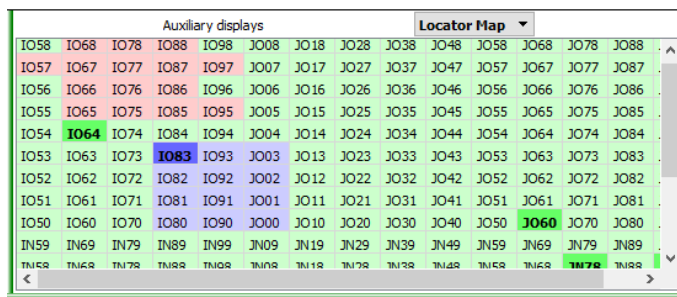


Figure 30 – Locator Statistics (Grid)

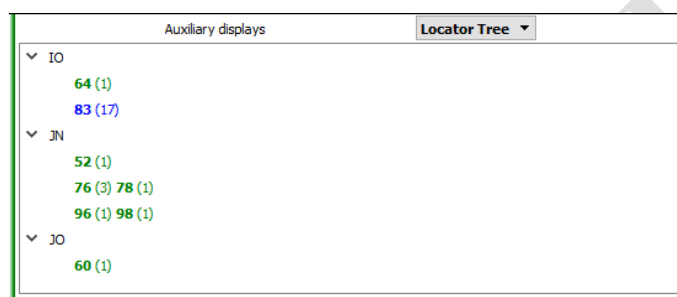


Figure 31 – Locator Statistics (Tree)

- Stats – displays information on best contact and scoring rates. The intervals for which the DX statistics are displayed can be modified by use of the P1 and P2 boxes at the bottom of the panel.

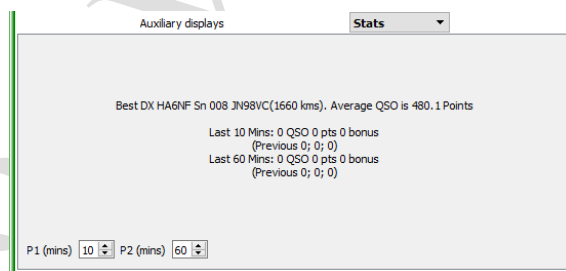


Figure 32 – QSO Statistics

3.2.3 Memories

This auxiliary panel can be used standalone as a ‘scratchpad’ or in conjunction with RigControl (see section 4.2.1) to provide the extended function described here. When first invoked there will be no memories present, they have to be created. The created entries are stored within the context of the currently open contest³⁰ and are preserved when a contest is closed.

²⁹ Since in the UKAC all squares now have the same bonus this is no longer really relevant. Grey indicates no bonus scheme in use. To get the colours shown, the B2 scheme would have to be in use.

³⁰ Hence are stored on the local machine, not any remote machine being used as a connection to the rig.

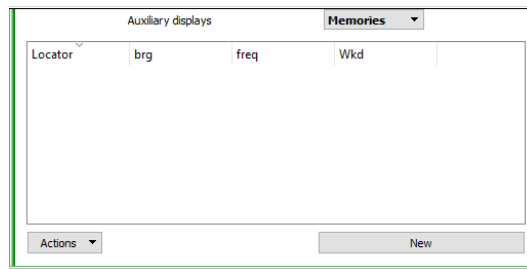


Figure 33 – Initial memories panel display

The order and size of the columns shown in Figure 33 can be changed using standard drag and drop mechanisms, and the display order can be controlled for each (ascending or descending). When entries are created the callsign appears in a new (autosized) column on the left hand side which cannot be moved or reordered (see Figure 35).

If “New” is clicked when the QSO entry panel has a Callsign and/or Locator present that data will be pre-loaded into the new memory (it can of course be modified), along with the current time³¹ (and bearing if available). If no data is present then an empty memory is displayed (although the time is filled in). See Figure 34 for examples. Clicking “OK” then loads the memory into the panel.

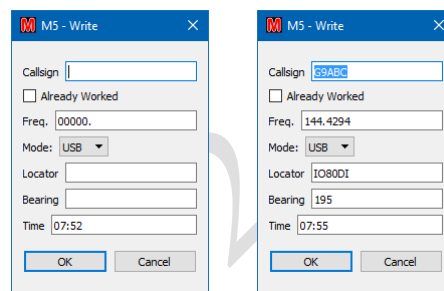


Figure 34 – Example memory content (blank and from QSO entry with RigControl active)

In order to save the memory some text is required in the callsign field; if nothing is supplied then “??” will be used, but beware when loading the contents back into the QSO entry panel.

The “Already Worked” box can be ticked if required which will then mark the memory entry in brackets in the same manner as any callsigns found in the log (see Figure 35)³².

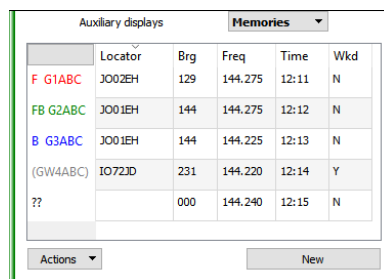


Figure 35 – Populated memories panel

³¹ The time information in a memory is lost if you close the contest or exit the logger.

³² This might for instance be used to mark a station that you want to work on another band.

If memories continue to be added then a scroll bar will appear once the available space is filled up. There is no practical limit to the number of entries, but they may be difficult to handle once scrolling starts.

The colour coding of the callsigns, and the preceding letters, indicate memories that ‘match’³³ the current rig frequency and the beam heading:

- Red F frequency within +/- 2 kHz.
- Blue B bearing within +/- 15 degrees.
- Green FB both frequency and bearing within the limits.

Worked entries (manually designated or detected in the log) are enclosed in brackets. The ‘??’ indicates a memory loaded with a blank (probably unknown) callsign in the QSO panel.

The “Actions” button provides a pull-down list of options (as shown in Figure 36) to Read, Write, Edit or Clear the selected memory or clear memories:

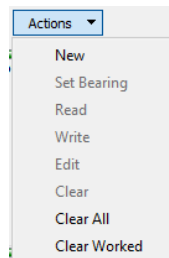


Figure 36 – Memory actions menu

Right-clicking an entry will provide the same list.

- **“Read”** will load the callsign and locator information into the QSO entry panel, as will a single click on the callsign field of a memory.
- **“Write”** prepares to load the memory from the current QSO fields. It can be manually changed before confirming with “OK”; the current time will always be transferred.
- **“Edit”** is very similar to “Write”. It opens the selected memory to allow the contents to be modified (without copying the QSO data).
- **“Clear”** deletes all contents of the selected memory (without confirmation).
- **“Clear All”** deletes all the stored memories (requires confirmation).
- **“Clear Worked”** deletes (without confirmation) all the memories that have been marked as worked as a result of being found in the log. Any entries that have been manually marked as worked are retained).

Memories may also be loaded by right-clicking on a match panel entry (see Figure 37).

³³ The match limits are currently hard coded.

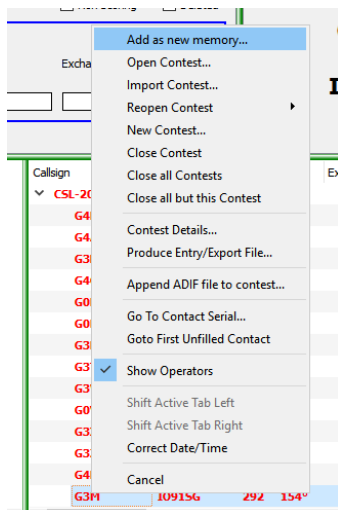


Figure 37 – Loading an entry from a match panel

A single click on the callsign field is equivalent to a read; a double click on anything but the callsign field is equivalent to edit³⁴.

Be careful not to (double) click the callsign if editing (or clearing) an entry (e.g. if a station's frequency has changed) because the information will load to the QSO panel and the rig will QSY to the old frequency if you are using Rigcontrol.

If a locator is present in the QSO box then the calculated bearing will be added to a new memory. If there is no locator then a manual bearing may be entered when a memory is created. If a bearing is manually entered and there is a locator present then the calculated bearing overrides the manual data.

3.3 Locator calculator

The Locator Calculator can be invoked from the main screen using “Tools/Locator Calculator” or via a button on the “Force” panel. If a contest is open the current station locator will be pre-loaded as Station 1.

Depending on the option selected, you can enter a Locator, a Latitude/Longitude, or a (UK) National Grid Reference³⁵ into the “Input grid ref” box (see Figure 38) and convert between the various displays. Selecting “Calc” (or hitting “Return”) will then calculate and display the other two formats. If both Station details are given it also calculates the distance and bearing from Station 1 to Station 2.

Use Exit, X, or Cancel to exit

³⁴ On the callsign it is the same as a single click, i.e. a read.

³⁵ Apologies to the Irish: the Irish National Grid requires a different centre meridian to allow for being somewhat west of the rest of mainland Britain, and hence is not calculated.

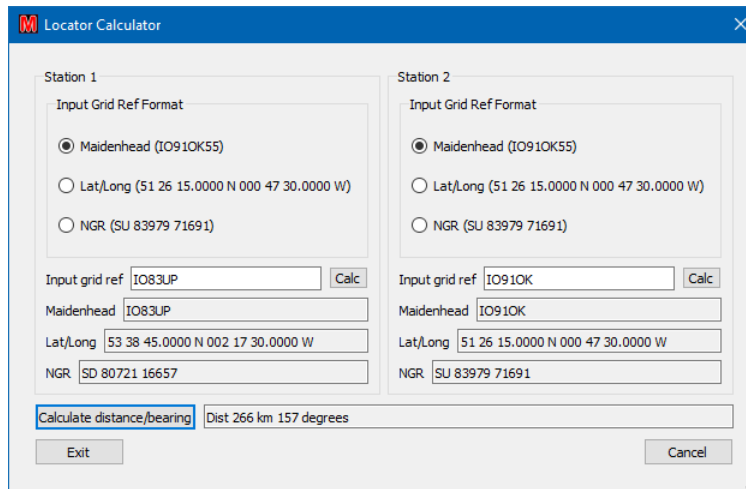


Figure 38 – Locator calculator

If the contest is set to “allow 4 figure locators” then the locator calculator will work out the closest point for a 4 figure locator e.g. JO88.

The distance is **always** calculated by the standard algorithm for the distance between the locator square centres.

Microwave locators using 8 figures such as IO80ST37 are supported. If the current contest/list has an 8 figure locator then the distance and bearing will be calculated to the accuracy of the 8 figure locators. During conversions, 8 characters always generated.

The precision used may appear excessive, but it is done in this way to enable conversion from NGR to Lat/Long and back again, and have some hope of ending up with the same result. Of course Locators are considerably less accurate.

But please note that precision doesn't necessarily imply accuracy; but those locations that can be checked accurately give a similar result.

If the entered location is too far outside the UK then no NGR is displayed.

3.4 Advanced operation

3.4.1 Multiple Contests

It is possible to have multiple contests open simultaneously. Each is displayed as a separate tab in the main logging screen as shown in Figure 39. It is possible to select/open multiple CSL (&LOG) files at once, but each will require acknowledgement. Contest sets (section 3.4.2) simplify the handling of multiple contests.

UTC	Callsign	RepTx	SnTx	RepRx	SnRx	Loc	brg	dist	Exchange	Comments
14:30	GW8IZR	59	008	57	012	I073TI	258°	142		
14:34	M0EEM	56	009	58	002	I082SQ	186°	108		
14:57	GM4PPT	57	010	59	010	I075SK	326°	244		
15:04	GM6JNJ	57	011	55	017	I075SO	328°	259		
15:13	G8NEY	58	012	59	018	I081VK	179°	246		
15:21	G8ONK	59	013	59	014	I083MR	282°	45		
15:31	M1BXF	59	014	59	037	J002BE	134°	230		
15:32	G3YDY	57	015	55	015	J001FQ	138°	286		
15:39	OH4WY	52	016	59	034	J011LC	126°	455		
15:44	2E0XT/P	59	017	59	018	I083TO	230°	8		
15:47	G4HGI	53	018	57	008	I083PL	236°	34		
15:59	M5BFL	57	019	56	045	I091OO	155°	249		

Figure 39 – Multiband contest

This was originally implemented to allow a single operator to handle a multi-band contest, e.g. the Christmas Cumulatives or all the microwave bands in a contest in the 432MHz-248GHz contest using a single copy of the program³⁶.

Separate scores, multiplier lists, etc. are maintained for each contest, but the current list of operators is shared.

3.4.2 Contest Sets

The above concept has now been extended to “Contest Sets”. These are managed via “Logs/Contest Sets/Manage Contest Sets” (see Figure 40).

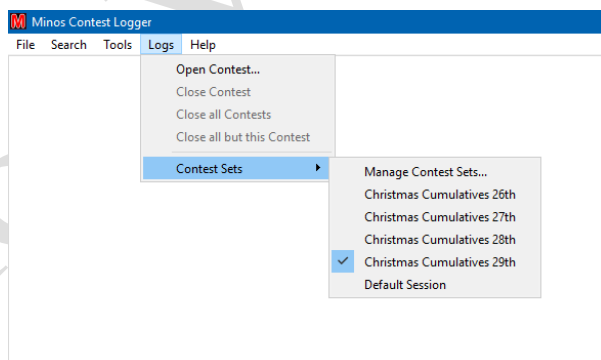


Figure 40 – Opening a Contest Set

Sets can then be created, logs added etc. When the programme is started all members of the highlighted set are opened³⁷. If this is for the first time there is no requirement to acknowledge each and every “Contest Details” panel as there would be if each was opened in turn (or as a block).

³⁶ Be careful! It is very easy to get confused and start entering data in the wrong contest log. This can be mitigated by use of RigControl (see “Quick Select” description in section 4.2.1).

³⁷ So be sure to leave ‘Default Session’ selected if you do not intend to be working with a set.

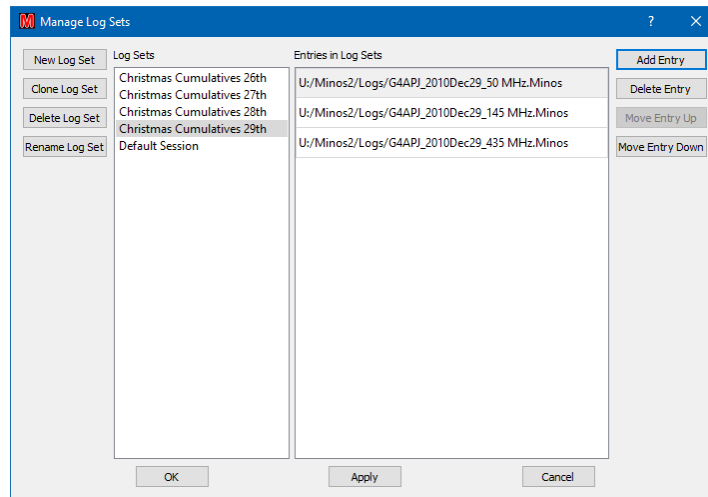


Figure 41 – Contest Set Management

Contest Sets can be considered as a single entity when opening/closing them. In the example above, selecting “Christmas Cumulatives 28th) would close the three logs from the 29th in one go and open all three logs for the 28th.

3.4.3 Archive Lists (CSL)

Warning: however good and accurate you think your list is, DO NOT TRUST IT and don't copy details from it that you have not heard on the air. Remember: people, and especially portable contest stations, move from time to time, or sometimes just work out their locator more (or even less) accurately. Not only that, but if you “invent” details that haven't been copied off air you are cheating, and if caught, you may well be disqualified from both that and other contests.

The format of a CSL file is described in section 5.1.3. It can be produced manually or the utilities detailed in Section 6 can be used to create the file and maintain the content.

CSL files are opened using “File/Open Archive list”. Once a list is loaded the file is closed, so that any changes to the file will only be picked up the next time it is loaded. Multiple files may be used which are controlled via “File/Manage Archive Lists”³⁸, that also allows the content of a file to be edited. In the event that a CSL needs to be removed this is also done via “File/Manage Archive Lists”.

Archive lists apply to all open contests

3.4.4 Time adjustments

Adjustments to the date and time may be made by selecting “Correct Date /Time” under the Tools menu option (see Figure 42).

³⁸ At least one CSL file must have been opened for this option to be active.

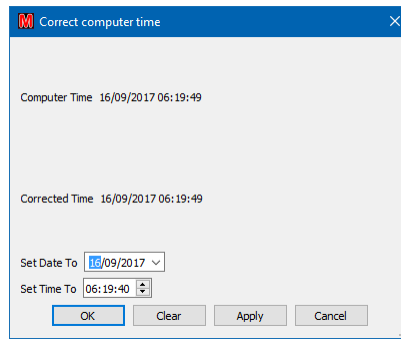


Figure 42 – Clock adjustment

3.4.5 Maintenance

The Contest Calendar should be updated at the very minimum on a yearly basis, preferably more regularly – especially in the first quarter when changes may still be in progress. This is done by clicking “File/New Contest”, “VHF Calendar”, “Download latest Calendar”

3.4.6 Hints, tips, and ‘tricks’

Reader contributions are welcomed here!

As with all software it is a good idea to take regular backups, but it is particularly important to save the configuration information before and after reconfiguring Applications – especially if you are doing an in-place upgrade and/or using a beta release. The internal format of the configuration files has changed several times, and there is no guarantee that it will not do so again!

Right clicking anywhere outside the match panels on the contest panel³⁹ allows a number of functions (see Figure 43). The first two groups of these are identical to the functions available from the File menu.

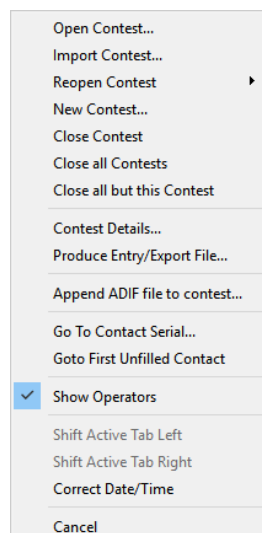


Figure 43 – Right click menu options on main screen

³⁹ Right clicking inside a match panel adds the option to add a memory – see Figure 37.

- **Go To Contact Serial:** invokes the QSO edit dialogue for the selected contact number.
- **Goto First Unfilled Contact:** opens the QSO edit for the first located contact with missing information (section 2.2.4).
- **Show Operators:** Controls display of the Op1/Op2 fields within the data entry panel.
- **Shift Active Tab Left/Right:** Move the currently selected tab in the specified direction. The tabs can also be dragged.
- **Correct Date/Time:** Correct the PC date/time (see Section 3.4.4)
- **Cancel:** exits the option panel with no action

3.5 Screen layout configuration

Clicking on “Tools/Screen Layouts” allows selection of the screen format to use for the active contest or, by clicking on “Configure Screen Layouts”, the definition of the formats to be contained in the list (Figure 44).

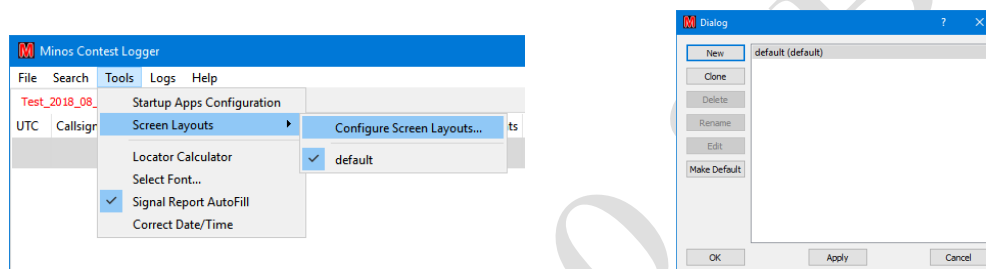


Figure 44 – Invoking the screen layout editor

A new layout is typically created by cloning the ‘default’ layout and giving it a name (Figure 45). A specific layout can be selected as the default for future contests by clicking “Make Default”. The layout used for an individual contest⁴⁰ can be changed using the “Screen Layout” pull-down list on the mid-right of the contest details panel (see Figure 10).

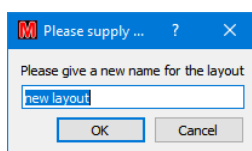


Figure 45 – Creating a newscreen layout

Clicking “OK” to a “Clone” operation or “Edit” to an existing (user defined) layout invokes a panel of the form shown in Figure 46 (the default layout) which allows the user to configure the content and placement of the Logger panels.

⁴⁰ The layout selected for a given contest is retained within the minos log file.

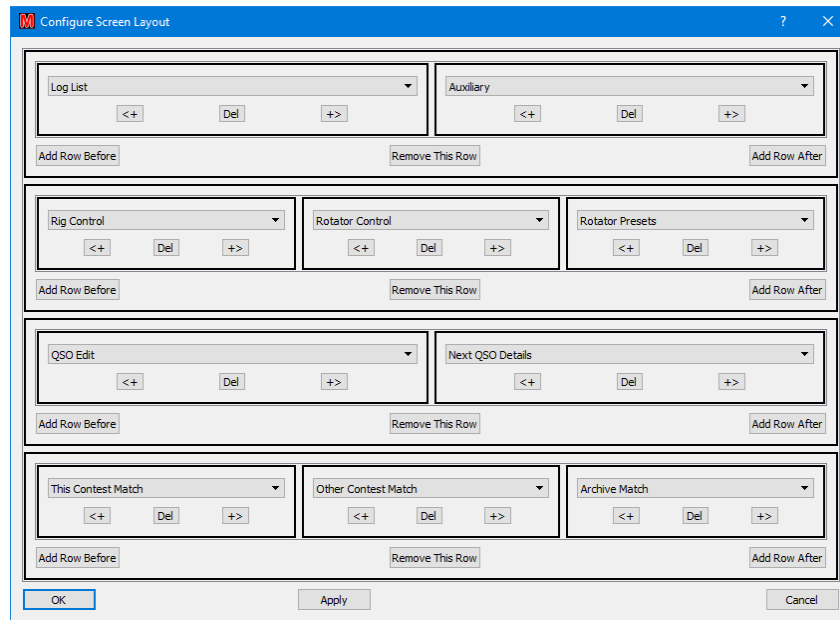


Figure 46 – Default screen layout configuration

Most of the buttons are self-explanatory, but for the sake of completeness a brief description of each is given here:

- “<+” and “+>”: Add a box to the left or right respectively of the current box. The content of that box is then selected from the pull-down list:

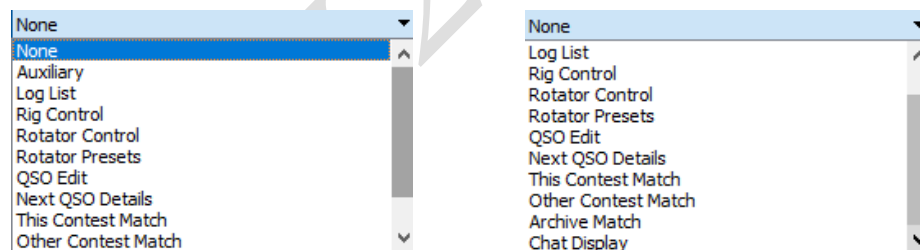


Figure 47 – Panel display options

- “Del”: Removes the panel from the display.
- “Add Row Before”/After: Inserts a single panel before/after the row in which it is clicked. Additional boxes can then be added to that row using “<+” and “+>”.
- “Remove This Row”: Deletes the whole row (multiple boxes if present).
- “OK”, “Apply”, “Cancel” options are available

In the event that all rows have been deleted⁴¹ an additional option “Add Row” will appear (see Figure 48) to allow rows to be recreated (or click cancel and start again).

⁴¹ Or none have yet been created in a **New** layout.

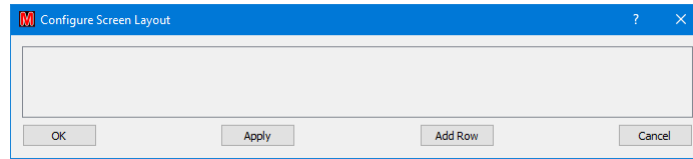


Figure 48 – Blank layout options

The Screen Layout configuration determines which panels are displayed where. To adjust the size of those panels slide the green bars (which change to red during adjustment) in the main Logger panel.

Note that (currently⁴²) panels do not appear/disappear as applications are configured, hence reconfiguration is desirable (if no RigControl/Rotator then you probably want to remove that row).

There are multiple ways of achieving the same effect!

⁴² This *may* change at some point in the future; meanwhile use the “Make Default” facility.

4 Integrated Applications

Some of the integrated applications described in this section are not yet available, even in rudimentary form, and may not appear for some time (if ever). As and when they do appear any functionality described here may have changed.

With 2.0.7 Pre-Beta there have been significant changes to the capabilities and configuration of (control) applications.

Two applications form the infrastructure which the functional applications use:

- **MqtServer** is fundamental to the operation of the applications. It is a message broker which runs on all machines where you want the applications. It handles all the message-passing between Logger, apps, and remote servers.
- **MqtAppStarter** is the Application Starter. It is used *instead* of the Logger when you want applications but not the Logger. This will generally only be used in complex multi-machine environments (see section 4.4.3.1).

Only one copy of these infrastructure applications should be run on any given machine at any given time.

4.1 Configuration

In order to operate correctly the application configuration must be defined. The following description is for the use of the Logger to set up the applications. Usage of the Application Starter is covered in section 4.4.3.1.

This configuration process is initiated either via the “Apps” button on the Welcome Screen or from “Tools/Startup Apps Configuration”^{43 44}. Either method will result in small “Minos App Configuration” screen which should be expanded from its default size before configuring anything (see Figure 49).

To add an entry click “New Entry” and select the required application from the “App Type” pull-down list. For simple ‘single user’ configurations all of the information required for the applications is auto-generated⁴⁵.

Since “Server” is required for everything else it is typically defined first. “RigControl” and then “Rotator” are the other most likely applications for simple (single machine) environments.

After applications are defined they are by default always all started. To avoid this use “Logger Only” in the startup splash screen or, on a more permanent basis, clear the “Enabled” box for the application in its definition, (remembering that Server is required to be active if **any** other application will be active).

⁴³ Reconfiguration is only allowed if the applications are inactive – use the “Stop Apps” button.

⁴⁴ In complex environments it can also be done using the MqtAppStarter application on a machine that is just acting as a server and thus not running the Logger.

⁴⁵ For details of more complicated configurations see section 4.4.

To store the configuration Click “Save and Close” (this will write the information to MinosConfig.ini and exit the window) or “Save and Start Apps” which will then require the “Cancel” button to exit the display.

By default “Hide Server Apps” is ticked for MqtServer to reduce the number of panels and icons displayed. Depending on configuration it may be advantageous to tick this for other applications.

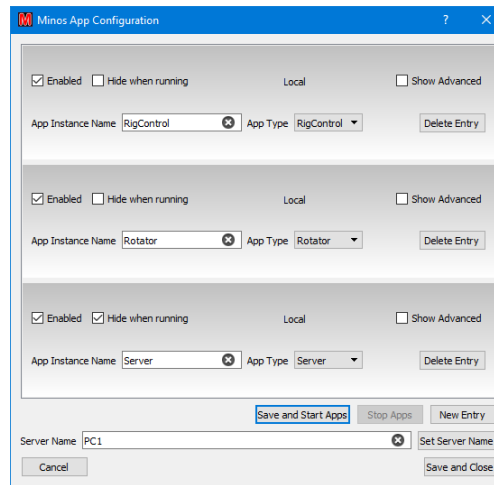


Figure 49 – Typical set of applications

“Delete Entry” can be used to delete erroneous information, after which “Save and Close” stores the changes.

When Server first runs, it requires Firewall configuration as shown in Figure 50.

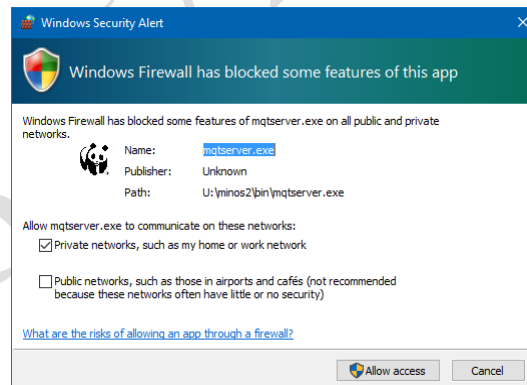


Figure 50 – Firewall message

If “Hide when running” has been unchecked for MqtServer then a window will appear that gives statistics on the active software components (Figure 51).

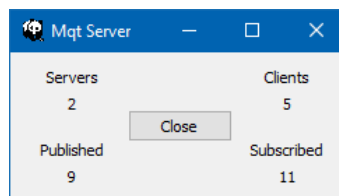


Figure 51 – Server statistics

When the Logger runs, a number of icons will also appear on the Taskbar for those applications that are not hidden. In Figure 52 the icons from left to right are:

- Main Logger
- Rig Control
- Chat (not displayed when running as a Logger panel)
- Rotator
- Monitor
- Server (not normally displayed)



Figure 52 – Icons on task bar

The control applications defined in this manner are then linked to individual contests using the pull-down lists from the Logger or from the Contest Details panel⁴⁶ (see Figure 53), either when creating a contest or by selecting “File/Contest Details”.

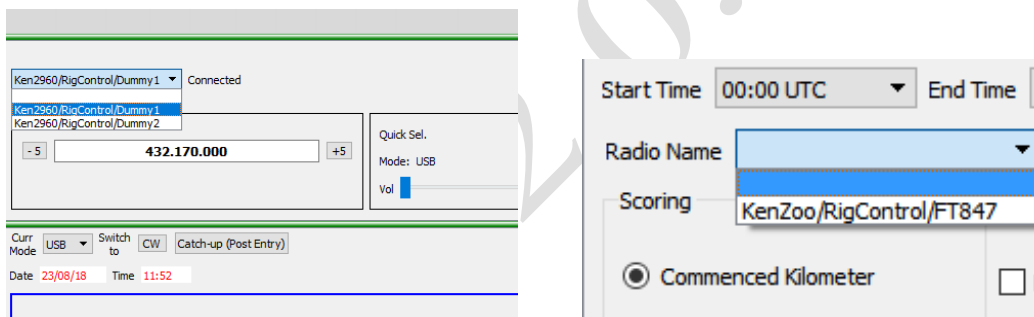


Figure 53 – Selecting a rig from Logger and Contest Details

4.2 Control Applications

These applications allow the control of rotator(s) and radio(s) using separate rotator and rig control applications communicating via (possibly USB connected) serial interfaces or, subject to finding someone who has one so it can be tested, full USB to their respective devices. The applications operate as servers for, and under the control of, the Logger.

While the programmes can control a range of rotator controllers and radios using the Hamlib library, it is not possible to test all of these devices as they are not available to the developers. Some controllers may not support all the commands available or a particular controller’s operation may not be fully understood.

A list of tested configurations for Rigs (and Rotators) is maintained at <https://minos.groups.io/g/users/files>

⁴⁶ Both examples are for rigs, the equivalent mechanism is used for rotators.

Feedback is welcomed on any information or changes that may be required to support as yet untested rotator and radio models.

The control applications can be connected to one or more contest log pages, selection of the rig in use (or rotator, if you have more than one) for a particular contest being via the drop-down lists as described above.

More complex arrangements such as different rigs for different bands are covered in section 4.4.

4.2.1 RigControl

This application connects to multiple rigs⁴⁷ using CAT interfaces via serial ports (which may be simulated via a USB device), or via direct USB connection if your rig supports it. The program will also support network radios, for example FlexRadio 6xxx. When you select a radio that supports a network interface, a network address and port number must be entered. **At present this facility is completely untested!**

It provides direct control of the selected rig and extends the functionality of the memory Auxiliary panel.

Before the Logger can connect to the RigControl application one or more rigs must be defined (so there is content in the pull-down list). Transverter configurations can also be defined (see section 4.2.1.1) to adjust the frequencies seen in Minos to reflect the actual transmit frequencies, rather than the driver frequency.

Selecting “**Configure/Setup Radios**” within RigControl invokes the ‘Radio Setup’ dialogue (Figure 54) that allows the radio(s) to be defined.

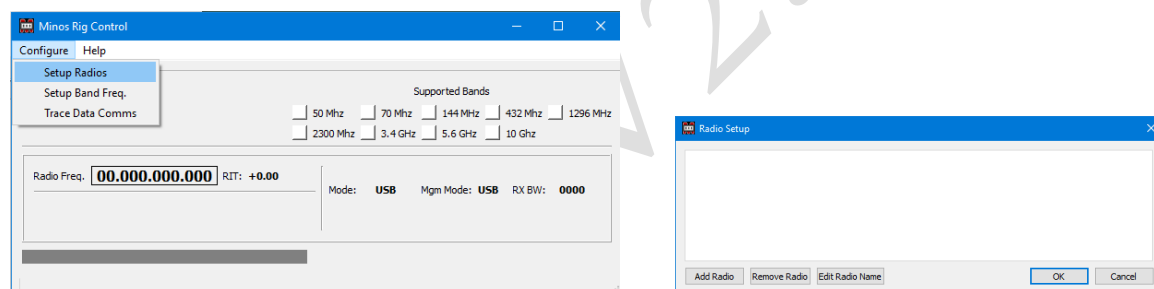


Figure 54 – Starting configuration of RigControl

Clicking “Add Radio” starts the definition of a rig (Figure 55).

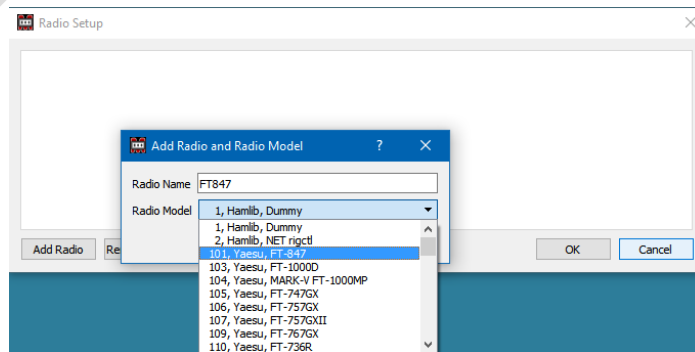


Figure 55 – Selecting the radio model

⁴⁷ More than any normal operator would want to use/can afford.

The “Radio Name” field is used with a contest to select (or display) which rig is currently being controlled (e.g. in Figure 61). Once defined, it may be changed using “Edit Radio Name” or removed (provided it is not currently active within the Logger) using the “Remove Radio” button.

“Radio Model” is a pull-down list containing an extensive range of rigs supported by Hamlib. Note that one of these is ‘Dummy’ which can be quite useful for testing/learning purposes.

Once the name and model are defined clicking “OK” continues to the setup of the interface to the device (Figure 56)

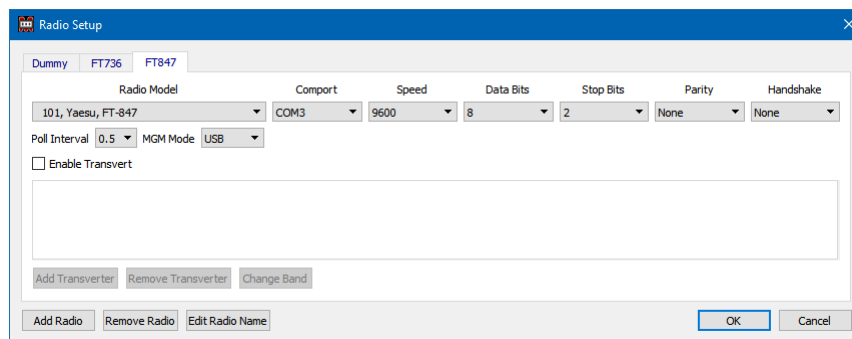


Figure 56 – Configuration of a serial connected rig

“Comport, Speed, Data Bits, Stop Bits, Parity” drop-down lists configure the serial communications⁴⁸ with the CAT interface of the radio selected from the “Radio Model” drop-down list.

“Handshake” Controls the flow control setting (e.g. XON/XOFF or CTS/RTS). Ensure this is correct for your interface (generally “None”). Some USB interfaces need CTS/RTS to be enabled to power the device.

If mis-configurations have occurred error messages such as “Please select a Com Port” may appear.

“Poll Interval” defines the interval in seconds at which RigControl interrogates the rig for information.

“MGM Mode” selects the rig mode to be used when MGM (data) communications are used.

“Enable Transvert” starts the transverter configuration dialogue, see section 4.2.1.1.

If your radio is an Icom then an additional “CIV” box will appear (Figure 57). Either leave the field blank to use the default CIV address of the radio, or enter the CIV number e.g. 0x66.

⁴⁸ Tip: check the settings on anything else already being used, e.g. WSJT-X.

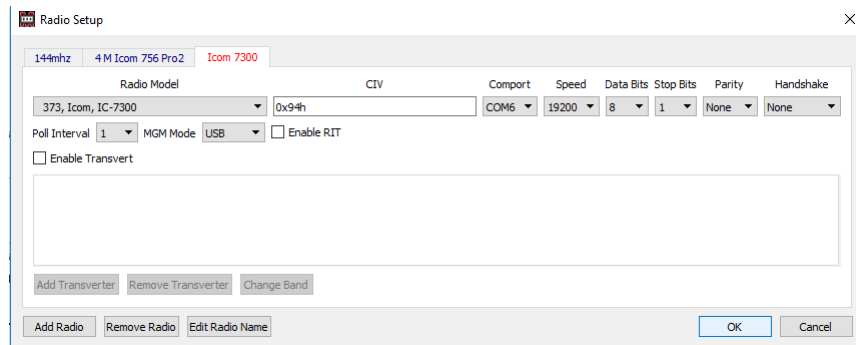


Figure 57 – Configuration of an ICOM rig

Note: With the IC7300 you need to ensure that "CI-V Echo Back" is "ON" in the settings⁴⁹.

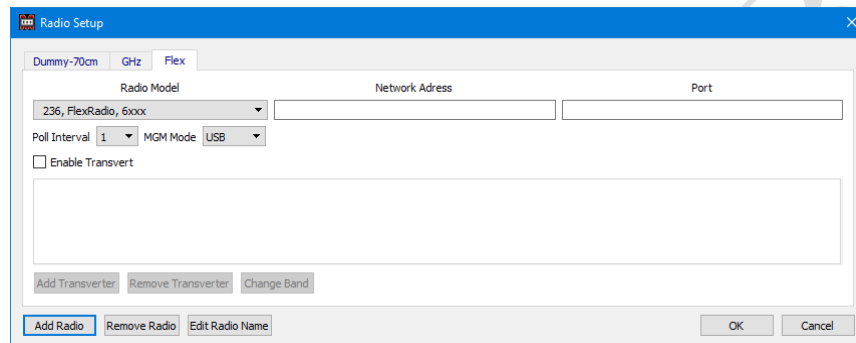


Figure 58 – Configuration of a network connected rig (FlexRadio)

A network connected radio (e.g. a FlexRadio) requires the target “Network Address” and “Port” to be entered (Figure 58).

If there are multiple rigs to be defined configure everything on one instance of RigControl and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

“**Configure/Setup Band Freq.**” (Figure 59) allows changes to the frequency the rig goes to when a contest is opened or when a “Quick Sel” band change is made from the Rigcontrol panel of the Logger (Figure 61). The default frequencies are chosen to avoid such things as calling channels. This option allows you to change the start frequency to your favourite start frequency on a band⁵⁰.

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

⁴⁹ All Icom rigs need Transceive OFF and USB Port Link to remote.

⁵⁰ It does not prevent you from specifying calling channels etc., so be careful.

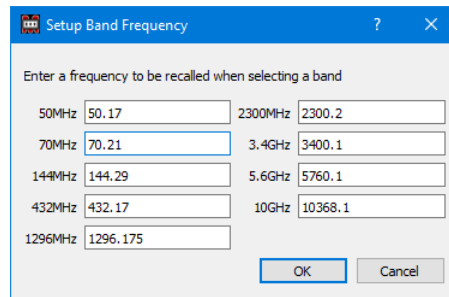


Figure 59 – Setup Band Frequency

“**Configure/TraceLog**” turns on the trace log (see section 5.3). It should be turned off (the initial default) in normal operation.

Once the configuration has been done, the radio to be connected to the Logger can be selected either via the Rigcontrol panel pulldown, which shows which radio is in use at any particular time, or the Contest Details boxes (see Figure 53).

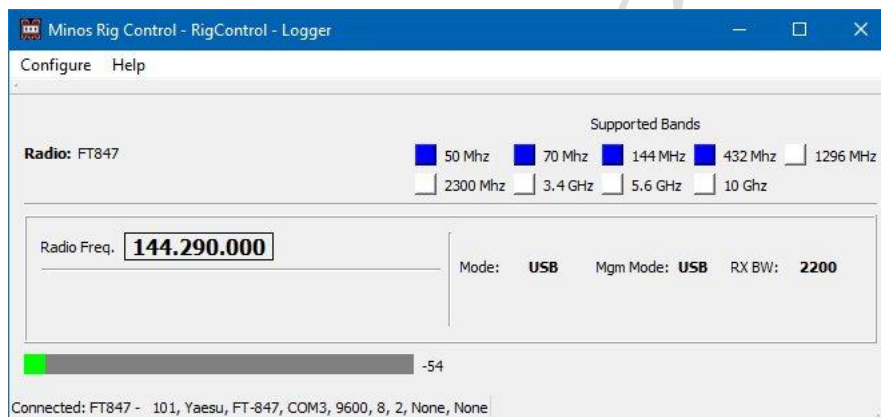


Figure 60 – RigControl operational

A status line at the bottom left of the panel (Figure 60) shows the Connected/Disconnect status⁵¹, and which radio has been selected, with what CAT configuration.

Note the “Supported Bands” boxes towards the top right. Blue indicates that a band is supported directly whereas yellow indicates the use of a transverter.

At the bottom of the panel there is an “S-meter” bar, although in the example the indicated magnitude is lower than S1.

Once configured to operate in conjunction with the Logger a contest may utilise the rig selected within the application, and the RigControl panel becomes visible in Logger (see Figure 61). As can be seen, because the application is connected to the Logger, it shows ‘Logger’ in the title, together with the name assigned to the instance of the app being used they are using (RigControl, by default).

⁵¹ Selecting a radio initiates connection.

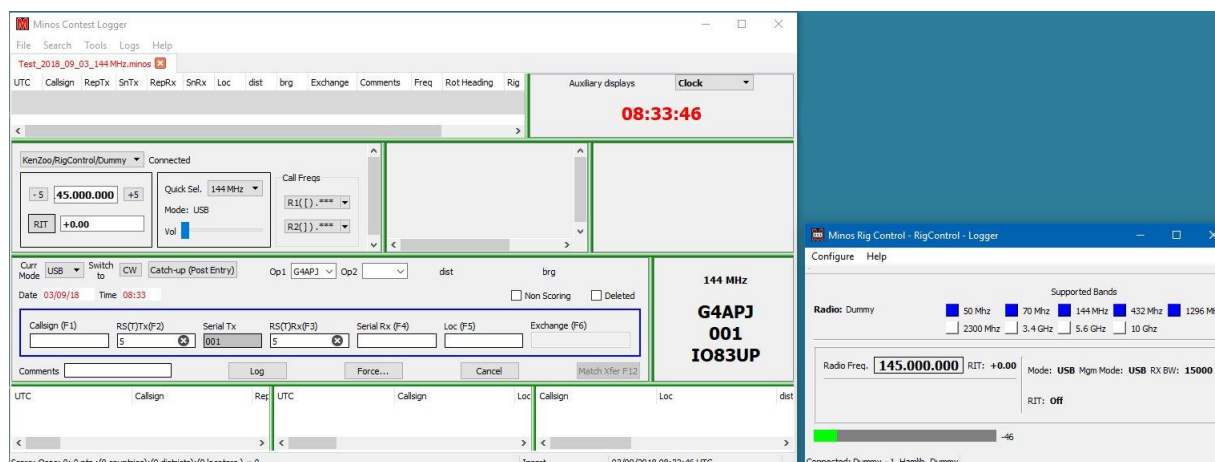


Figure 61 – RigControl connected to Logger

Within the RigControl panel (see Figure 63) on the top left is the name of the selected radio and its status which should normally be ‘Connected’ for correct operation⁵².

The current frequency⁵³ of the rig is displayed and may be changed by selecting characters inside the box and either typing in characters or using the mouse wheel to increase/decrease (band limits are enforced). If the mouse wheel is used then the changes are made immediately; if characters are type then “Return” causes the rig to be updated.

The border will turn red to show it is in edit mode.

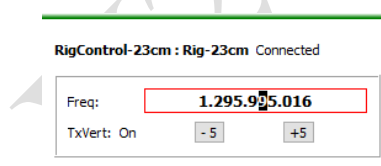


Figure 62 – Frequency change in progress

Ctrl-f selects the KHz digit, or use the mouse to select digit, as the tuning step. With the cursor in the “Freq” box you can now use the mouse wheel or the up/down arrow to tune. Left/Right arrow change the selected digit, hence the step.

Hit “Enter” to select the new frequency or escape to cancel. The frequency may be ‘nudged’ using the -5/+5 buttons to allow calling near the frequency of a station being received⁵⁴.

If the selected rig supports RIT control via CAT, and RIT is enabled in the Radio configuration, then the fine tuning information will appear under the main frequency display as in Figure 61. Similarly, if the rig supplies volume control information this will be added to the display under the Quick Select list/Mode display.

⁵² If the target of a remote connection is not active when log started then it loses the connection and the target is not in the list. It reconnects automatically when the target becomes available.

⁵³ The calculated frequency is displayed if a transverter is in operation.

⁵⁴ Useful to call a station near a frequency that is in use.

In multi-band contest with multiple tabs the Logger initially displays the current rig frequency on all tabs using a particular RigControl, switching to the frequencies from the “Band Frequencies” list as contest tabs are selected (or “Quick Select “changed).

During operation any changes to that frequency are only propagated to the active tab; those not in focus do not get updated.

When an inactive tab is selected and put into focus the frequency (and mode) used the last time the tab was active is retored to the rig (whatever band that happens to be, i.e. if the 2M tab has a 6M frequency from the startup then selecting the 2M tab will cause the rig to go to a 6M frequency).

“Quick Select” provides a quick method to get the rig to the right band for a particular tab at startup, as an alternative to physically changing the rig controls⁵⁵. The frequency comes from the “Band Frequencies” list.

The current mode (USB/CW/FM/MGM) on the radio is displayed⁵⁶, and is changed as selected within the QSO data entry frame (since it may be changed in the Logger when RigControl is not in use).

The rig frequency is adjusted to compensate for changes between USB and CW.

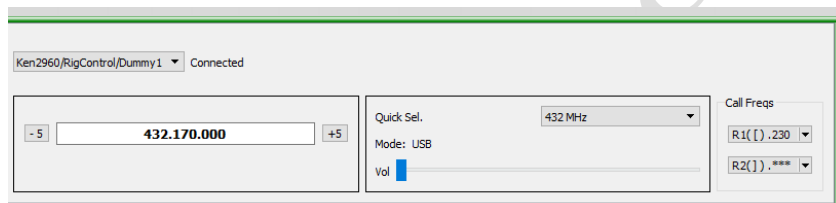


Figure 63 – RigControl panel within Logger

When RigControl is in operation the memory functionality described in section 3.2.3 is fully realised, as well as the R1/R2 button functionality described here. These buttons represent the ‘run frequencies’ and mode selected by the operator and are configured using the same R/W/E/C dialogues as the standard memories (see Figure 64).

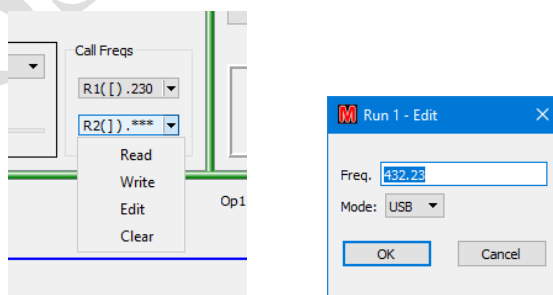


Figure 64 – Storing a run frequency

The button displays the frequency if set and “***” if nothing is stored. In Figure 63 R1 is set to (432.)290⁵⁷ and R2 is empty. The square brackets indicate the CTRL shortcut symbols that can be used. Run memories are held within, and retained with, each log.

⁵⁵ Restarting the Logger clears the tracking, all tabs revert to the current rig frequency at startup.

⁵⁶ At startup the radio is always switched to USB

Under some circumstances (most likely comms failures) errors may be returned from the Hamlib routines controlling the rig (Figure 65). If this occurs use “X” to exit RigControl, which will then restart and should re-establish communication. If it does not, then investigate the comms settings and cables.

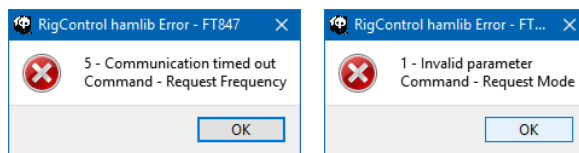


Figure 65 – Hamlib error messages

<<Looking at ways of coping with this on remote servers>>.

4.2.1.1 Transverter Configuration

If “Enable Transvert” is ticked the rig being defined/modified will be used with a transverter and the buttons are enabled to allow this definition (see Figure 66), including making the “Enable Transvert Switch” option visible.

“Add Transverter”, “Remove Transverter”, and “Change Band” are used to control the definition of possibly multiple transverters/bands.

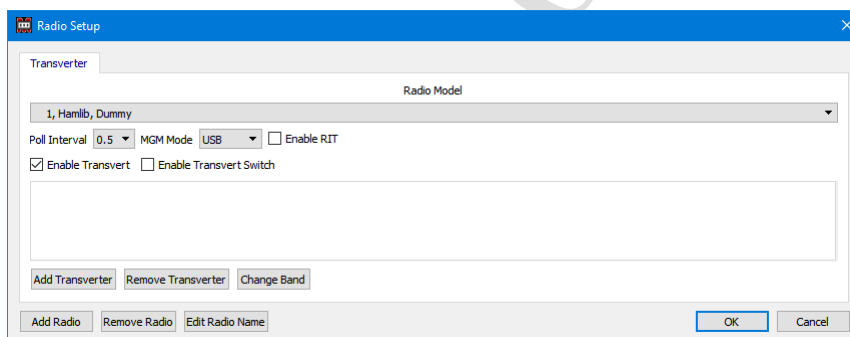


Figure 66 – Transverter configuration

When you add a transverter, a list of bands are displayed (Figure 67). The radio may support the band directly, but a transverter can still be selected for that band if for instance its RX performance is better than the radio.

Beware: The 2300MHz band consists of three segments: 2300-2302 MHz, 2310-2350 MHz, and 2390-2450 MHz. Please take this into consideration when using this facility.

If the radio supports a band for which a transverter is configured, the transverter frequency combination will be used in preference to the radio band. Configure a different radio definition (or don’t configure a transverter for that band) if this is not what is wanted.

⁵⁷ It is presumed to be 432.290; if you have changed bands it will still be on the original band.

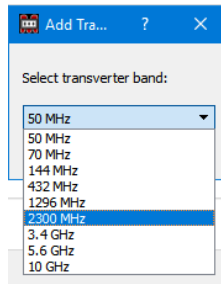


Figure 67 – Transverter band selection

Having created a transverter tab for one or more bands (Figure 68) the frequency definitions govern how it operates.

“**Radio Freq**” is the frequency the rig actually tunes to get the desired output

“**Target Freq**” is actual operating frequency

“**Offset Freq**” is the difference between them, which the transverter produces⁵⁸

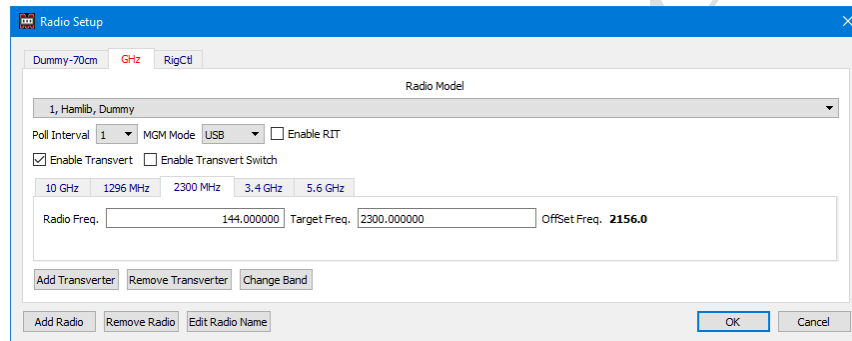


Figure 68 – Transverter definition

These frequencies are then reflected in the Rigcontrol display and passed to the rig. So, with the above definitions for 2.3 GHz if 2,300.1 MHz is entered in the Rigcontrol panel of the Logger the Rigcontrol display will appear as in Figure 69 and a frequency of 144.100 will be sent to the rig.

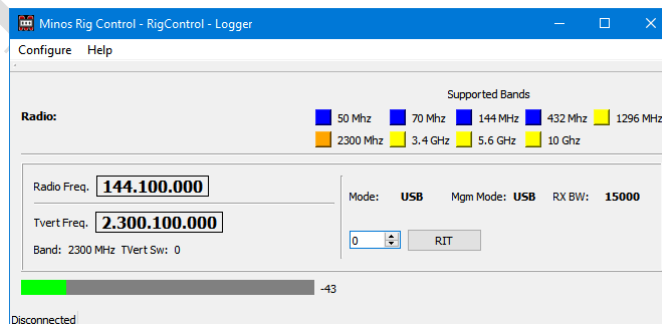


Figure 69 – Transverter in Rigcontrol display

⁵⁸ Currently the focus must be moved from the entry boxes to get the calculation displayed.

There is the facility to drive external switching of the transverter either via an external com port or via a network message to an external control system⁵⁹

“Enable Transvert Switch” enables this facility for all defined transverter bands if ticked. It also makes the “Enable Local Switch Control” box and “Transvert Switch Number” dialogue visible.

Regardless of the path used, the message to be sent is entered in Transvert Switch Number box. It is free format, which might typically be just a simple number, and is ‘wrapped’ in a ‘:’ as to start and ‘LF’ to terminate.

“Enable Local Switch Control” displays the pull down list to select the com port to be used (Figure 70). The message transmission is then via the selected com port. The ‘standard’ default set up is assumed (9600baud, 8bit, 1 stop bit, no parity, no flow control

The intention is that the message will be read by an Arduino or a Pi based ‘switch box’ (see Appendix section 7.4) .

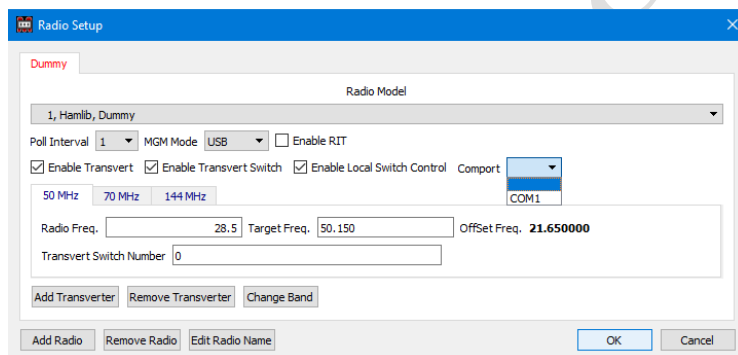


Figure 70 – Defining Local Switch Control port

4.2.2 Rotator

The application has many similarities to RigControl. It can connect to multiple rotators⁶⁰ (from those supported by the Hamlib library) via serial ports (which may be simulated via a USB device), or via direct USB connection if your rotator controller supports it. The application is intended to be used under control of the Logger.

Once the application has been connected to a particular contest (see above) the rotator(s) in use need to be configured so that the selected controller operates under control of the Logger. Figure 71 shows the startup with Rotator defined to a contest but with no controllers yet defined.

Selecting “Configure/Setup Antennas” allows the rotator(s) to be defined.

⁵⁹ Still under development, but see section 7.4.

⁶⁰ Once again, more than any normal operator would want to use/can afford.

Figure 73 shows an example configuration of a rotator.

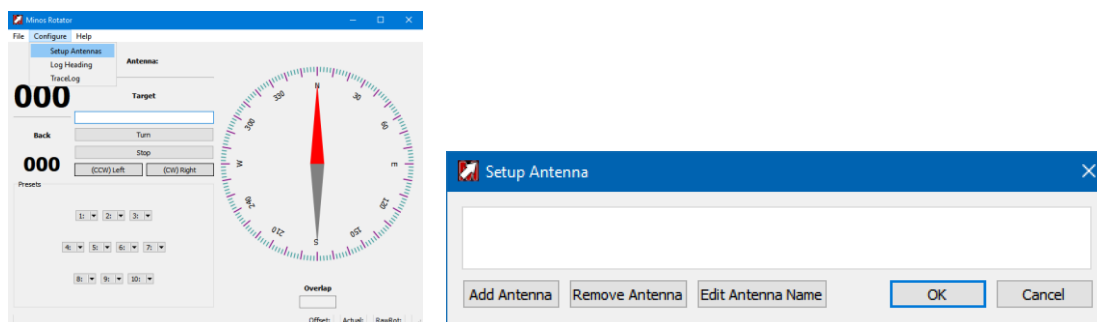


Figure 71 – Starting configuration of Rotator

Clicking “Add Antenna” starts the definition of a rotator (Figure 72).

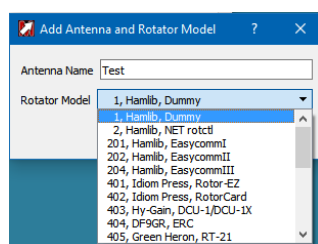
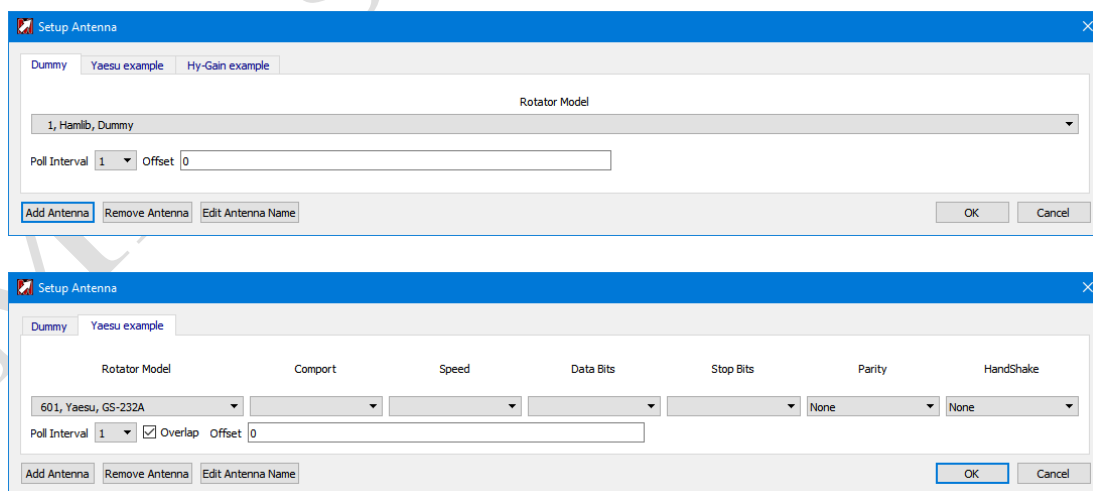


Figure 72 – Selecting the rotator model

The “Antenna Name” field is used to select (or display) which rotator is currently being controlled (e.g. Figure 79). Once defined, it may be changed using “Edit Antenna Name” or removed (provided it is not currently active within the Logger) using the “Remove Antenna” button.

Depending on which rotator is selected different options will appear (Figure 73).



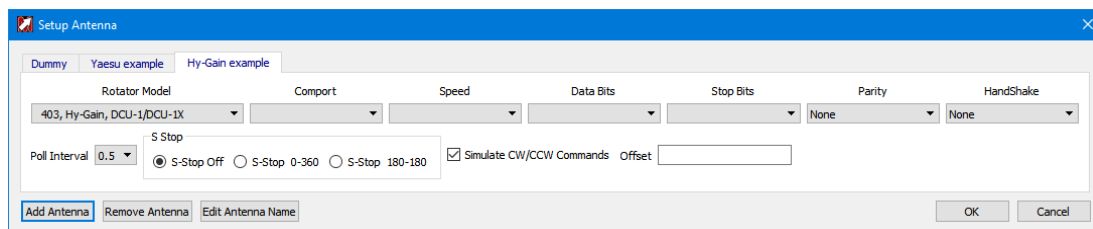


Figure 73 – Defining various rotators

If there are multiple rotators to be defined configure everything on one instance of Rotator and then restart the Logger (and thus applications). This will ensure that the definitions are visible to all instances.

The “Antenna Name” is the method by which the particular rotator is selected from the pull-down list.

“Rotator Model” is a pull-down list containing an extensive range of rotator models supported by Hamlib. Note that one of these is ‘Dummy’ which can be quite useful for testing/learning purposes.

“Comport, Speed, Data Bits, Stop Bits, Parity” drop-down lists configure the serial communications with the rotator controller selected from the “Rotator Model” drop-down list.

“Handshake” Controls the flow control setting (e.g. XON/XOFF or CTS/RTS). Ensure this is correct for your interface (generally ‘None’).

“Poll Interval” controls the interval between polls of the rotator; the default of ‘1’ should be correct for most installations.

“Overlap” will only appear, and will be ticked by default, if the Rotator being controlled supports overlap capabilities. The capability for overlap and the amount of overlap allowed is derived from information in the Hamlib library.

Some rotator models do not support overlap and the rotator module in Hamlib may not support overlap either.

The box should only be unchecked⁶¹ to turn off overlap when using an overlap-capable protocol with a rotator that does not support overlap. The rotator will then only turn between 0 - 360 degrees. If Overlap is unchecked then the Overlap box will not be displayed on the Rotator Display.

Beta testers are required to check out this functionality.

The “Offset” to support an antenna mounted offset from the rotator bearing a value between +90/-90 may be entered in the “Offset” box⁶².

Rotator bearing is always shown in Actual on status line. With the offset (from Rotator definition) the pointer indicates the 'real' beam heading, not the rotator current setting.

⁶¹ If Overlap is unchecked and an antenna is sitting in an overlap region, you may not be able to rotate the antenna. The antenna will need to be manually moved < 360 or > 0

⁶² There is also an antenna offset field in contest details to allow for any skew between antennas in the same stack, or inaccuracy in the direction setting of the antenna.

With some rotators such as the Hy-Gain in Figure 73 (or if “Overlap” is unchecked on Yaesu rotators⁶³) a number of settings appear to allow a rotator with a north-stop to be mounted with the mechanical stop to the south while the sensor is sending 0 – 360, instead of -180 to +180:

“S-Stop Off” the rotator will act as normal

“S-Stop 0-360” Minos inverts the reading of 0-360 that the controller sends

“S-Stop 180-180” is used for rotators such as some old Stolle models, that are mounted with a south stop, but use a compass sensor that sends 180 - 180⁶⁴.

“Simulate CW/CCW Commands” is for rotators that don’t support turn-right/left commands in their protocol. When the turn right/left buttons are pressed, a ‘rotate to end stop’ bearing command is used. If that causes a problem unchecking the tick box will cause the turn right/left buttons to be removed.

“Configure/Log Heading” allows a location and filename to specified for logging bearings when “Logging Enabled” is ticked (see Figure 74). Bearings will continue to be appended to the specified file⁶⁵ every time Minos Rotator is started until “Logging Enabled” is cleared. The “Bearing Difference” is the incremental change required before a new bearing is logged.

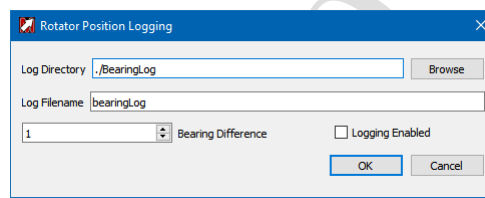


Figure 74 – Configuring Rotator Position Logging

```
Mon Oct 30 11:33:57 2017 Bearing is 19 Degrees
Mon Oct 30 11:33:57 2017 Bearing is 22 Degrees
Mon Oct 30 11:33:58 2017 Bearing is 25 Degrees
Mon Oct 30 11:33:59 2017 Bearing is 26 Degrees
```

Figure 75 – Example Bearing Log

“Configure/Trace Data Comms” enables the tracing of communications between Minos and the Radio⁶⁶. It is recommended that this is only turned on for diagnostic purposes. If checked the state is saved and it will be enabled when RigControl is restarted, so remember to turn it off when done. See section 5.3 for more information on trace logs.

⁶³ The Yaesu protocol is probably the most commonly emulated.

⁶⁴ Actually the stop is set to 181 to 179.

⁶⁵ If you want a new file, either specify a different name (saved through restart) or move/rename the existing file.

⁶⁶ Standard trace messages are now recorded at all times.

Once the configuration has been done, using “Antenna Select” to specify the rotator in use connects the application to the Logger. The rotator in use can be changed by selecting a different entry from the pull down list; the top left of the Rotator panel in the Logger will indicate which is being used (see Figure 79). The current status of the rotator (e.g. “Turning CW”) is also shown adjacent the connection information.

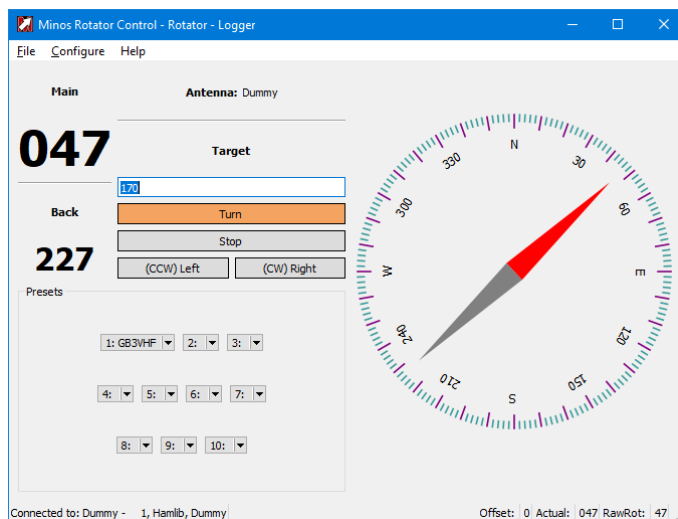


Figure 76 – Rotator control operational

A status line at the bottom left of the panel shows the Connected/Disconnect status, and which rotator has been selected.

“Configure/Edit Presets” allows the definition of the ten memories visible in the bottom left of the Rotator panel (see Figure 76). Each consists of a name (typically a callsign) and the relevant target bearing. These memories are effective for **all** rotators on the host machine, and are retained across restarts⁶⁷.

The presets are replicated in the Rotator Control panel of any Logger contest accessing the rotator⁶⁸.

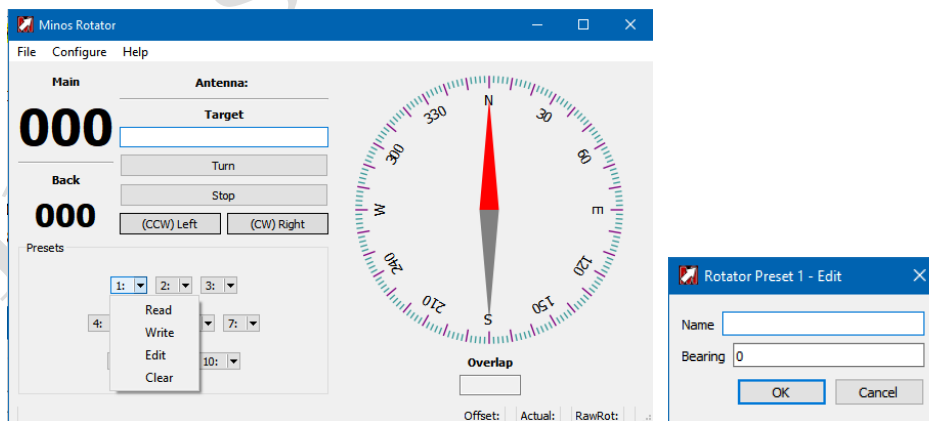


Figure 77 – Configuring Rotator control pre-sets

⁶⁷ Entries may be removed by going into “Configure/Edit Presets” and deleting the individual content.

⁶⁸ On any machine accessing the host in complex configurations.

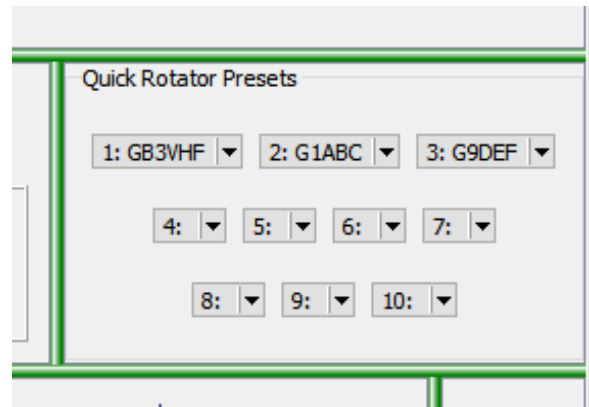
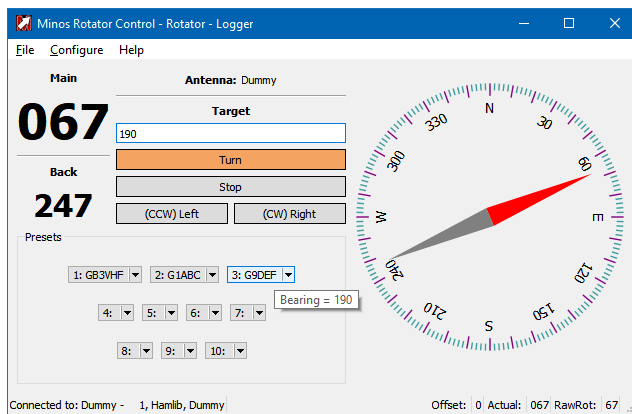


Figure 78 – Configuring pre-sets

Clicking one of these entries initiates rotations to the requisite bearing.

Once configured to operate in conjunction with the Logger a contest may utilise the rotator selected within the application, and the Rotator panel becomes visible in Logger (see Figure 79). As can be seen, because the application is connected to the Logger, it shows ‘Logger’ in the title, together with the name assigned to the device currently selected to be used.

With Rotator in operation there are enhancements to the memory functionality described in section 3.2.3. The memory content is extended to include bearing information. When a memory is read the bearing is inserted into the Target field for the rotator and rotation can be initiated by “Turn”/”Return”/ “Ctrl-T”.

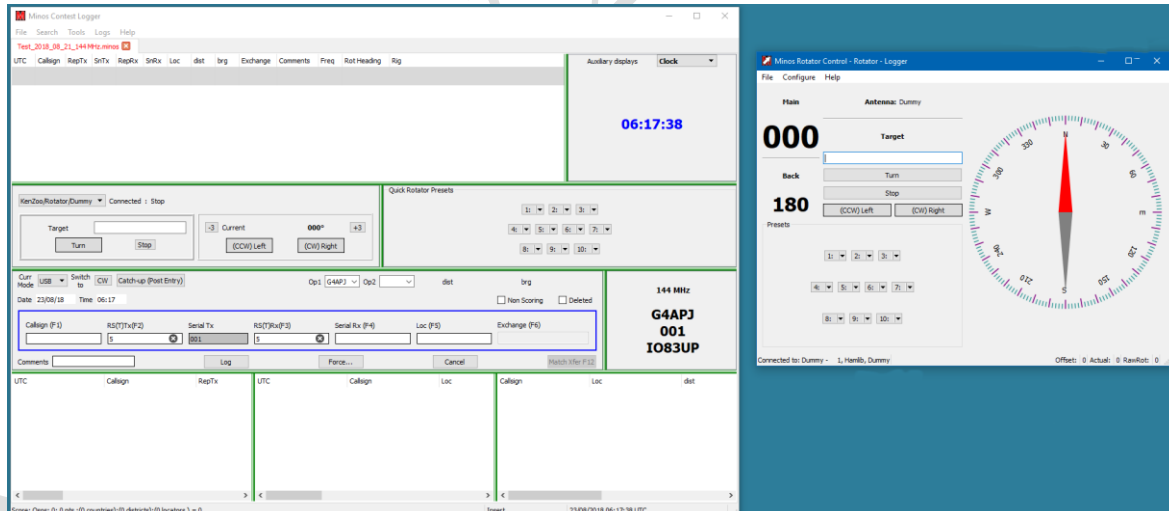


Figure 79 – Rotator connected to Logger

The antenna can be controlled from the Rotator application or from the buttons in the Logger. If actions are initiated from the Rotator application the panel in Logger shows what is happening.

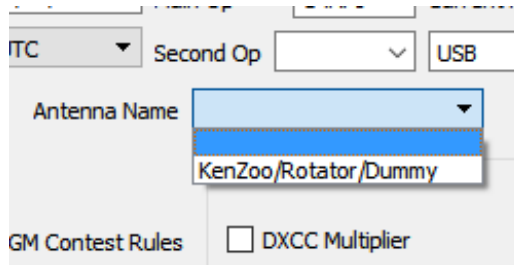


Figure 80 – Selecting the Rotator Application Instance

Within the Rotator panel in Logger (see Figure 81) the text at the top left shows the rotator in use ('Dummy2' in this example) and the status (e.g. 'Turning to bearing').

The left hand box shows the target bearing ('170') and the "Turn" button changes colour to show that an operation is in progress (it clears when complete). The rotation to the requested bearing will occur via the shortest valid movement using overlap.

The right hand box shows the "Current Heading"⁶⁹ for the rotator in use as well as the control buttons "(CCW) Left" and "(CW) Right"⁷⁰ and the 'Nudge buttons' (-/+) which nudge the rotator 3 degrees every time the button is pressed⁷¹ (in the direction of the adjacent box, i.e. + is CW).

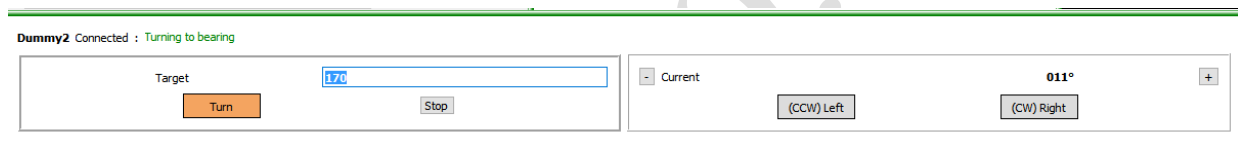


Figure 81 – Rotator panel within Logger

The "Target" (heading) may be entered manually, or is calculated from any QRA locator entered in the data entry panel in the Logger. This locator itself may be derived from an entered callsign found in the archive and transferred using F12 (or equivalent).

If a target bearing has been manually entered then rotation may be started by hitting RETURN, else click the Turn button or Ctrl-t to start rotation. Rotation may be stopped by clicking the "Stop" button on the left or by using the shortcut (Ctrl-s).

While rotating the bearing is checked for end stop, if reached a stop will be automatically issued.

Logging a contact while the rotator is turning logs the 'incomplete' bearing (i.e. the bearing at the moment of logging).

The Rotator application (see Figure 76) displays the bearings (Main and Back), together with a 'compass bearing' schematic of the current bearing heading (actual, calculated using any offset defined). A status line

⁶⁹ Red indicates overlap > 360 degrees and blue for overlap < 0 degrees.

⁷⁰ Left and Right are from the perspective of physical buttons on the rotator.

⁷¹ If a button is selected when rotator is moving, a Stop command is issued before the selected command is issued.

at the bottom left of the panel shows the Connected/Disconnect status and the name assigned to the rotator that has been selected.

There are control buttons to “Turn”, “Stop”, “(CCW) Left” and “(CW) Right”.

Some rotator controllers do not support a Rotate Left (CW - ClockWise) or Rotate Right (CCW - Counter ClockWise) Command. To allow the CW and CCW buttons to function, the command rotate to minAzimuth (typically 0 degrees) is sent for CCW command and the command rotate to maxAzimuth (typically 360 degrees) is sent for CW. The CW and CCW buttons are latching, selecting the buttons again will stop rotation. Rotation will also stop when the rotator reaches the endstop if a stop command is not sent first.

If a control button is selected while the rotator is moving, a stop command is issued before the selected command is issued.

If you change any value in the Current Selected antenna, you will be prompted to reload the values. You can choose “No” to skip and the values will be loaded when the program is restarted.

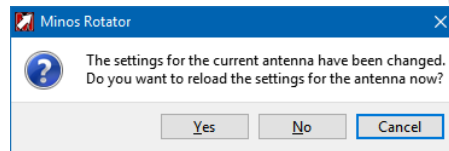


Figure 82 – Rotator settings changed

4.3 Information Applications

4.3.1 Chat

Chat is similar to an IRC application allowing text messages between stations. It can now be run in either (or both) of two configurations: as a separate window or, with suitable Screen Layout definition, as a panel in the Logger.

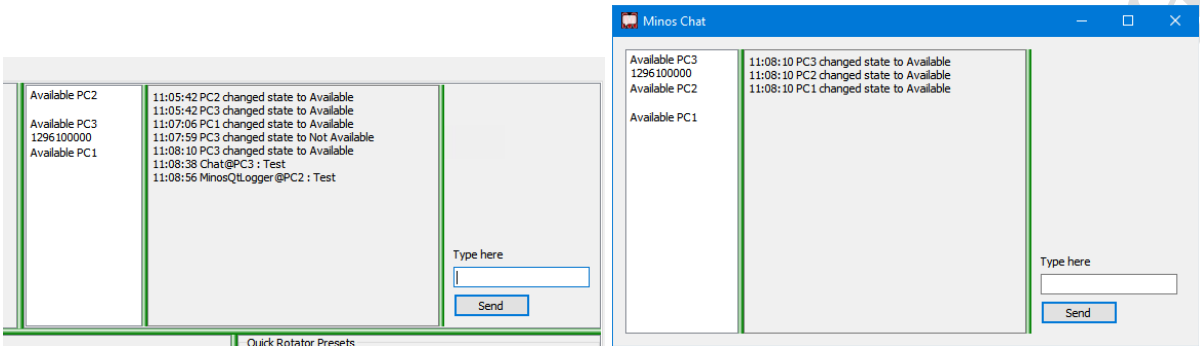


Figure 83 – Examples of Chat usage (Logger Panel and Standalone app)

The application is self configuring as machines come Available (chat application is visible) or go NotAvailable (chat is not visible, machine may or may not be active). To send a message (to all available stations) enter it in “Type here” and click “Send” (or hit “Return”).

4.3.2 Monitor

<<Probably also changing to be another contest panel at some point>>

Monitor allows display of the logs on both remote and the local machine.

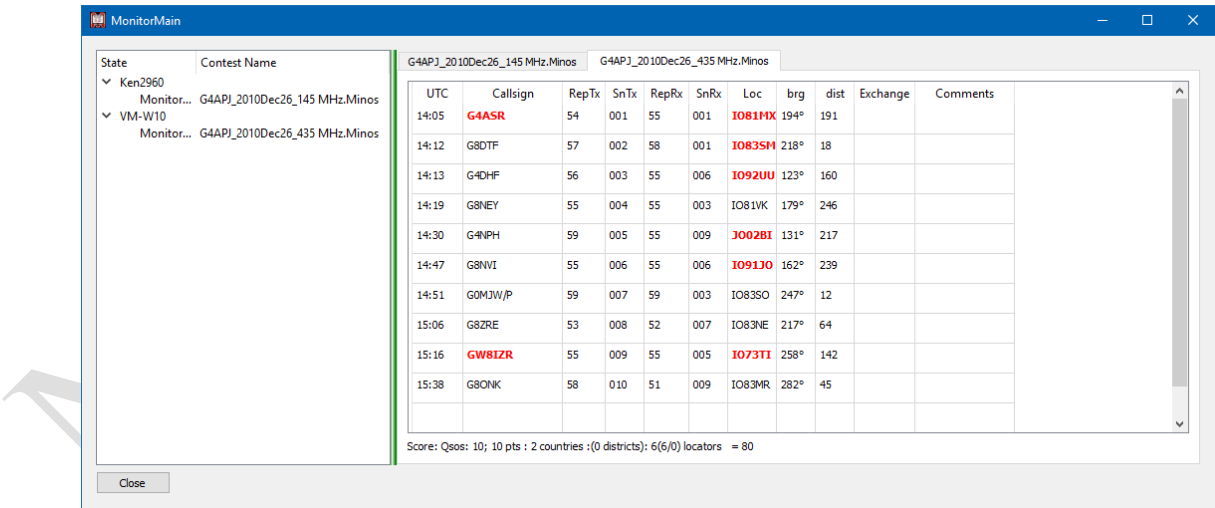


Figure 84 – Example of Monitor usage

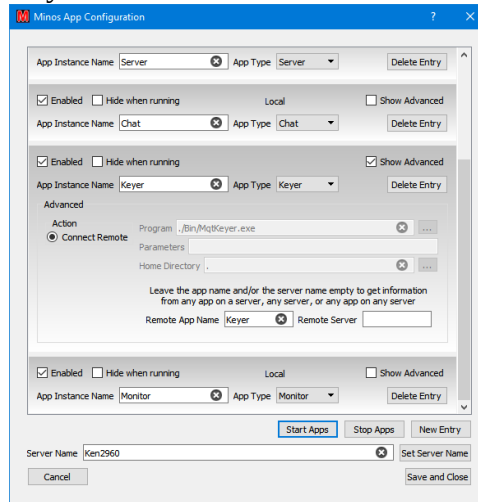
The application is self configuring as Loggers come online/disappear and contests are opened/closed. The contests on a particular machine may be expanded/hidden using the chevrons on the left.

To start monitoring a contest double click its entry in the left hand panel. A new tab will be open, display data from the monitored log. Click “X” on the tab to terminate monitoring of that contest log.

4.3.3 Keyer

<<TBA>>

Can only be defined to operate remotely.



4.3.4 DXcluster

Will be a window that can connect and display the spots, but the functionality is still to be decided.

Configuration
Usage

4.3.5 Band Map

Another future facility that is planned is Band Map. It requires Rig Control to be working first and hence will take a bit longer.

Configuration
Usage

4.4 Advanced Configuration

Advanced configuration is required for several possible activities:

- Development and testing
- Multi-PC environments necessitated by limitations on space, ports, or PC performance
- Multi-operators

Advanced configuration is invoked by clicking “Show Advanced” on one or more applications within App Configuration (see Figure 85).

Complex/multi-machine environments involve other components and careful planning is required, particularly of the names used.

The defaults are quite sufficient for single machine operation with multiple Rigs or Rotators attached, with manual selection of the one being used at any particular time.

4.4.1 Development/debug

For development/debugging it is a simple matter to change the “Program” invoked for a particular function. Use the selection button on the right to browse to the development/test version of the application; the “Parameters” passed to it and the “Home Directory” used may also be modified.

It may be necessary to change the “App Instance Name”⁷² (“Rotator-debug” in Figure 85) in the “Apps Configuration” in order to link the contest being used for the testing with a particular (debug/development) application.

Some test scenarios may require the use of a remote machine, and thus may be considered in the same manner as complex/multi-machine environments

⁷² The App Instance Name must be unique on any one machine.

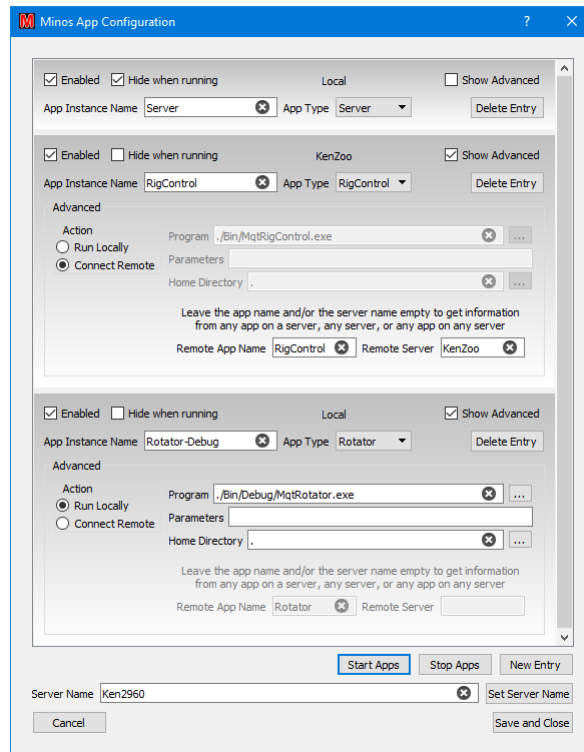


Figure 85 – Advanced configuration

4.4.2 Multiple rigs

The rigs defined through “Configure/Setup Radios” are common to all instances of RigControl on any given PC. A number of factors may force the use of a multi-machine environment:

- Number of available ports on any one PC
- A single PC is not powerful enough to handle everything
- The number of concurrent contests, hence:
 - Operators
 - Available monitors
 - Contest logs open
 - Active instances of Minos (only one per machine)
- Physical separation

For information on configuring multiple machines see section 4.4.3

Within each contest that uses a specific rig, change the Contest Details “Radio Name” definition (or use the pulldown list in the Logger) to select the requisite rig; the result is stored within the contest, but can be changed at any time.

Configuration information is shared between all instances of RigControl on a single machine. If the wrong entry from the “Select Radio” pull-down list was inadvertently selected it would cause problems that might be difficult to diagnose (especially under contest pressures)⁷³.

⁷³ There is a degree of protection due to contention for the comms port(s).

The same technique would apply if multiple rotators were in use (but changing the Rotator App names obviously), and of course both techniques can be used together.

4.4.3 Multi-machine environments

In multi-machine environments all the machines have to be in the same LAN broadcast domain; no intervening routers.

The server name for a given PC defaults to the machine name, but can be changed by setting the new name in the “Server Name” box at the bottom of the App Configuration panel and clicking “Set Server Name”.

The default instance of RigControl controls the locally connected rigs, and acts as the target for remote access to the local systems.

In order to have different rigs on different machines, multiple instances of RigControl must be defined; to avoid confusion these instances should have meaningful names. Probably the easiest is to append the server name⁷⁴ to which the instance connects, e.g. Rotator-PC2, or the functional area, e.g. Rigcontrol-VHF. Similarly, the target band could be appended, e.g. RigControl-2M and RigControl-70cm as shown in Figure 86, although this implies that each target PC only handles one band (possibly through multiple machines).

The choice will depend on the requirements/topology of the target environment, but make sure the planning is done before the contest, the result is documented and tested, and the final list is adhered to⁷⁵!

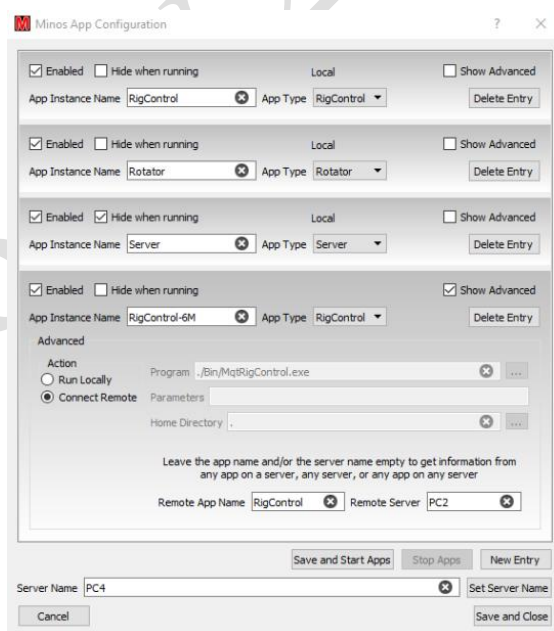


Figure 86 – Configuring two rigs, using remote RigControl

⁷⁴ Which need not be the same as the PC name, if changed through the Apps Config panel.

⁷⁵ You can, if you want, leave the “Remote App Name” and/or “Remote Server” fields blank within a ‘remote’ Rigcontrol definition. This will result in **all** instances becoming visible and possibly resulting in confusion, so once again; plan it!

It might be useful to stagger the configuration process; get one working before doing the next.

“App Instance Name” is the name by which the connection is known on the local machine. “Remote App Name” is the name on the target machine, typically the default. Consistent naming helps – especially in more complex configurations (see section 4.4.4).

If the app is set to “Connect Remote” then “Remote App Name” and “Remote Server” must match the definitions on the remote machine. It is probably desirable to change the names on both machines to indicate that the app is remote/being used by a remote machine.

Obviously the name of the server on which the instance runs must match the name defined in that machine’s server definition.

<< **Beware:** there is nothing to stop multiple contests sharing a rig or rotator; any such access must be coordinated!>>

4.4.3.1 AppStarter

If running a machine with no Logger (e.g. a Pi, acting just as a server device) then the Application Starter is used *instead* of the Logger to control the applications.

If the server machine is a Windows PC then the Application Starter can be started using the command

```
>start /d C:\Minos2 C:\Minos2\Bin\MqtAppStarter.exe
```

(within the system startup file) to ensure the correct default directory is used. AppStarter itself does not display much information (Figure 87) but the configuration dialogue is the same as the Logger EXCEPT that appstarter has an "autostart" checkbox.

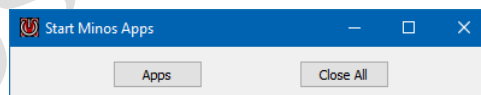


Figure 87 – AppStarter

If the server machine is a Raspberry Pi then Application Starter is run from the login script, invoked by having the machine auto-login. It then runs headless, with any access through SSH.

4.4.4 Example complex configuration

This configuration is intended to show the methods and capabilities rather than reality; it is overly complex!

Analyse requirements for a particular situation (cable lengths?)

Figure 88 shows an example is a sort of ‘super contest group’ combining VHF NFD with the 432MHz-248GHz contest, and a possible way such a hypothetical situation might be configured:

- PC1 Runs the Logger for 6M and 4M with a rig for each band, both connected to and controlled by PC2. It has a single rotator with two Antennas, one for each band, serving the two contests.
- PC2 Runs the Logger for 2M, which can use either of two rigs - one of which is controlled by PC4. It has two rotators, one of which is controlled by PC4. It also runs the 6M & 4M rigs for PC1.
- PC3 Runs the Logger for 70cm ‘and up’ (23, 12, 9, 6, 3), with a separate rig for 70cm (controlled by PC4) and locally controlled rig/transverters for the other bands. It has two rotators; the one for 70cm is controlled by PC4, the locally controlled rotator handles all the antennas for the GHz bands.
- PC4 runs the AppStarter to control the 2M rig/rotator from PC2 and the 70cm rig/rotator from PC3.

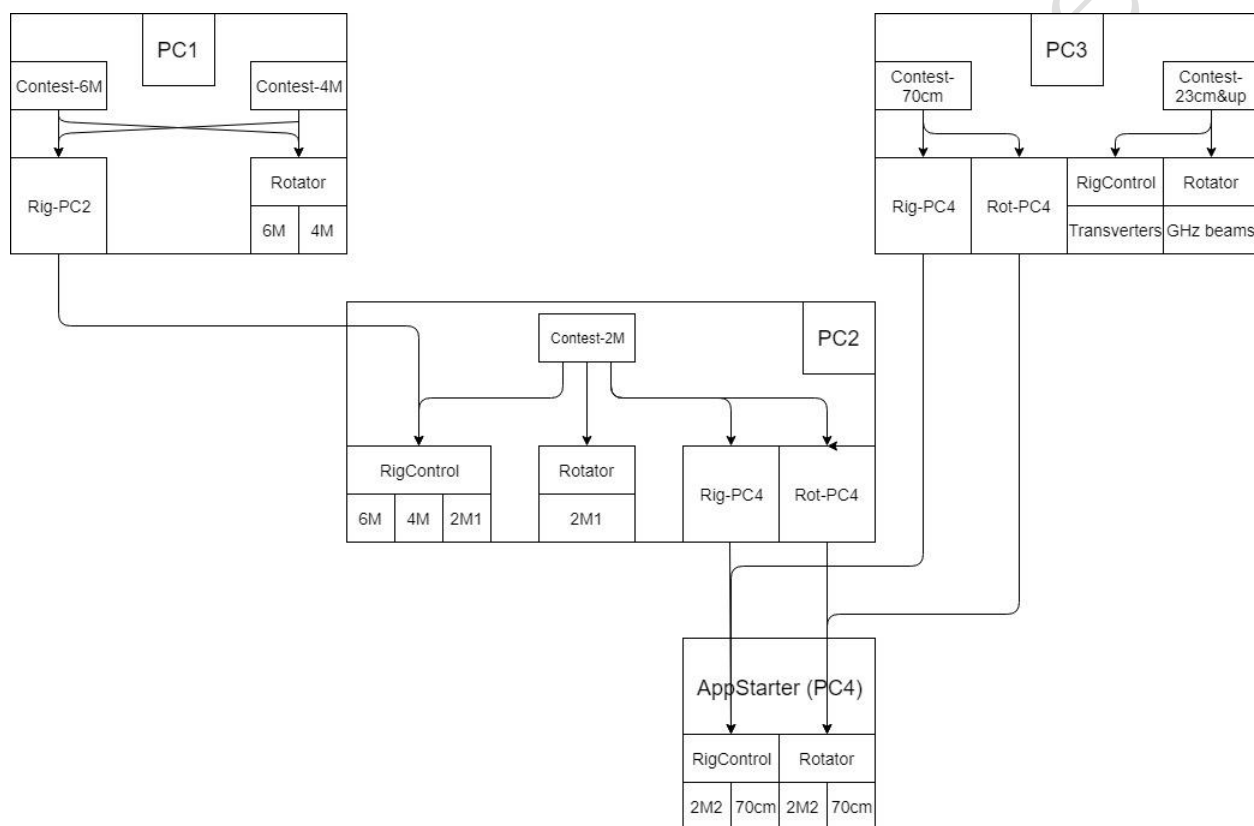


Figure 88 – Example complex configuration.

An example naming plan is shown in Figure 89

	Log	Appname	Connects to
PC1	6M 4M	RigControl-PC2	PC2/RigControl/6M PC2/RigControl/4M
		Rotator	6M & 4M aerials
PC2	2M	RigControl	6M, 4M, 2M1 rigs
		RigControl-PC4	PC4/RigControl/2M2
		Rotator	2M1 aerial
		Rotator-PC4	PC4/Rotator/2M2

PC3	70cm	RigControl-PC4	PC4/RigControl/70cm
	23cm-3cm	RigControl	23cm-3cm rigs (transverters)
	70cm	Rotator-PC4	PC4/RigControl/70cm
	23cm-3cm	Rotator	23cm-3cm aerials
PC4	(AppStarter)	RigControl	2M2 & 70cm rigs
		Rotator	2M2 & 70cm aerials

Figure 89 – Naming plan.

5 Internals

This section is still very much work in progress.

Minos is written using the open source Qt framework. If you want to get involved, then ask on the GJVLit reflector. Details of how to build the software are documented in the main source repository.

When you do anything that makes a log change Minos appends a record to the file with the changes; one of the fields is a QSO sort key so that records can be correlated. The exact details are not documented here; anyone who really wants to 'go inside' it can look at the code (or ask G0GJV).

There should be no requirement to edit the file; indeed, it is highly recommended that you do **not** do so, in case of corrupting it.

5.1 Folders and files

The overall folder structure is:

- **Bin:** programmes and DLLs (MqtLogger.exe is the main Logger programme)
- **Configuration:** configuration and INI files
- **Lists:** the default location for CSL file(s)
- **Logs:** the default location for storage of the .minos log files. They are not particularly interesting reading in raw format; certainly *don't even consider* editing them!
- **Tracelog:** diagnostic traces are written to this folder

5.1.1 Configuration files

- **Bandlist.xml:** describes the amateur bands and the various formats possible. This file should not be edited!
- **Cty.dat:** the DXCC countries and prefixes list. Documentation of the CTY.DAT format for CT9 can be found on the K1EA website (<http://www.k1ea.com/>). CTY.DAT should be used **unchanged**; any modifications required should be made to CTY.SYN. This file is updated at the same time as downloading the calendar⁷⁶.
- **Cty.syn:** As countries keep adding to their valid prefixes this file may need to be modified to keep up to date. Each line starts with a primary DXCC prefix, followed by synonym prefixes, separated by commas. You can use # on the start of a line to signify comments.
- **District.ctl:**
- **DISTRICT.SYN:**
- **Prefix.syn:** allows postcodes to be checked for correct country and also duplicate checking if a station changes country in the middle of a contest.
- **vhfcontests.*:** contest calendar(s) downloaded from the RSGB web page.

5.1.2 INI files

⁷⁶ The URL is <http://www.country-files.com/>.

The content is fairly self-explanatory, but if you want detailed information then look at the Open Source code. Some of the INI files are only created when program first run or as various applications are configured.

In general there should be no need to touch any of these files directly. The information is provided for interest/ completeness only.

It is strongly recommended that these files are not edited, use the provided interface to make changes!

“File/Options”

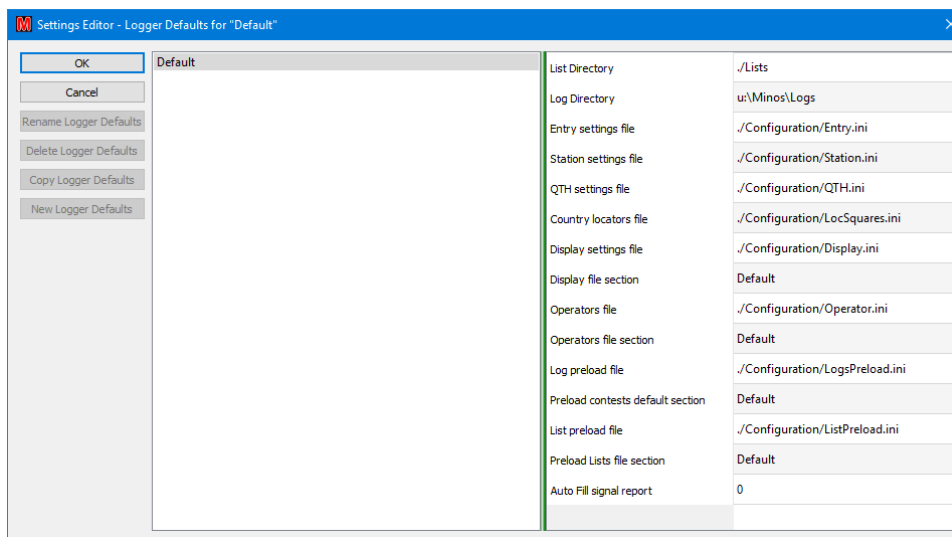


Figure 90 – Options Panel

\Configuration

- **AppConfig.ini:** the template for application definition.
- **Display.ini:** holds the selected statistics options and screen layout information.
- **Entry.ini, QTH.ini, Station.ini, Contest.ini:** contain the various bundles of information used to define a contest entry (as described in section 1.2).
- **MinosConfig.ini:** defines the configuration of the applications (see Section 4).
- **MinosLogger.ini:** holds a pointer to the logs location and the autofill setting.
- **ListPreload.ini:** contains the list of CSL files to be read on startup.
- **LocSquares.ini:** maps locator squares to countries to help validation.
- **LogsPreload.ini:** details the Contests and Contest Sets to be loaded on startup.
- **MinosRigControlConfig.ini:** tracelog and MGM mode control.
- **MinosRotatorConfig.ini:** the memory pre-sets from Rotator.
- **Operators.ini:**
- **UKACBonus.ini:** definition of the bonuses to be used for various squares within UKAC.

\Configuration\Antenna

- **AvailAntenna.ini:** which antennas are available, and the settings to access the rotators.
- **RotatorCurrentAntenna.ini:** the currently selected rotator

\Configuration\Radio

- **AvailRadio.ini**: which rigs are available, and the settings to access them.
- **FreqPresets.ini**: the list of initial frequencies for each band
- **RigControlCurrentRadio.ini**

\Configuration\Radio\Transverter

- **<RadioName>TransVertRadio.ini** : the definition of the badns and settings for the transverter(s)

5.1.3 CSL format

Archive lists (.CSL) are now proper CSV files, and the fields may be quoted, plus there is now a possible extra field in the list for “number of contacts” with the call/locator/exchange combination.

```
# Any line with a hash as the first character
# is considered a comment and is ignored
# Format is
# <Callsign>,<Locator>,<Exchange>,<Count>
#
2E0BMO,IO83PO,WN,
2E0CNJ,IO83TO,,
2E0DOD,IO83XI,,
2E0DOD/M,IO83XI,,
2E0DOD/P,IO83XG,,
2E0LAE/P,IO93BJ,SD,
2E0NEY,IO81VK,,
2E0RFX,IO83WO,OL,
2E0UAC,IO92FJ,,
2E0UOG,IO83PN,WN,
2E0XLG/P,IO83WV,,
2E0XLG/P,IO84WC,BD
...
```

The ‘count’ field is the ‘number of contacts’ which is used in various ways by the archive list maintenance tools.

5.2 Matching Rules

Matches are evaluated with an embedded space character representing any character (single character wild card). Matching is soft - if no exact matches are found, the program will progressively soften the match criteria - first, remove any /x (suffix, e.g. /P) from the callsign, then ignore the locator and QTH fields, and finally ignore the callsign prefix and number (e.g. G0GJV/P will be looked for as GJV). At each stage, if any matches are found the softening stops. The match display varies - for the current contest it is the same as the main display; for other contests it loses irrelevant information - date/time, reports and serial numbers, giving more room for more important information. As the match process progresses the header of the match list will indicate progress. Typing any character in these fields will kill the current match process and restart it.

In addition to the above match process the callsign is subject to a rigorous duplicate check. A duplicate (if present) will always appear first in the match list.

5.3 Trace Logs

Trace logs can be very useful for diagnosing problems. In the event of finding a problem please try to supply a detailed list of steps leading up to the fault, and the appropriate trace log.

Rigcontrol and Rotator log trace messages all the time apart from those passing over the comms link to the transverter(s).

Trace logs are pruned by age - 10 days

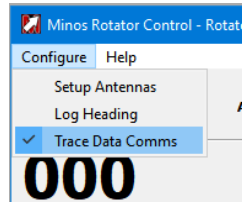


Figure 91 – Enabling extended trace log for Rotator

Tracelogs can be generated for diagnostic purposes. This can be enabled from the Configure menu where the TraceLog line can be enabled or disabled by checking the menu item. It is suggested that this should be disabled in normal operation.

Example tracelog with current antenna settings⁷⁷.

```
09:28:07.731 *** Antenna Updated ***
09:28:07.731 Rotator Name = Test Antenna
09:28:07.731 Rotator Model = 603 Yaesu,GS-232B
09:28:07.731 Rotator Number = 603
09:28:07.731 Rotator Comport = COM8
09:28:07.731 Baudrate = 19200
09:28:07.731 Databits = 8
09:28:07.731 Stop bits = 1
09:28:07.731 Handshake = 0
09:28:07.731 Antenna Offset = 0
09:28:07.731 Current Max Azimuth = 450
09:28:07.731 Current Min Azimuth = 0
09:28:07.731 South Stop Flag = 0
09:28:07.731 Overrun flag = 1
09:28:07.731 Rotator Max Baudrate = 9600
09:28:07.731 Rotator Min Baud rate = 150
```

5.4 Registry entries

⁷⁷ These settings can also be obtained via “Help/About Rotator Config” in the Rotator application. RigControl has the same capability.

All registry content is collected under [HKEY_CURRENT_USER\Software\Minos2Qt]

Don't even consider editing it, even if you think you understand it. It can be helpful to reset defaults when re-installing the programme for test purposes. This may be done by using the option on the file pull-down list.

Minos V2.0.8 Beta

6 CSL Management Utilities

6.1 Archive Utilities (G4AUC)

These four utilities can be used to create and populate archives (CSL files), check for possible errors in archives and logs, and merge archives. They keep track of the number of times (and the dates when⁷⁸) a station has been worked. All use standard browse mechanisms to locate (or create) the files being processed.

Download “setup_ArchiveUtilities_2.4.0.6.exe” to a convenient location on your computer from <https://sourceforge.net/projects/minos/files/> and then run it.

It uses the standard install dialogue to install the utilities to a default location of “C:/Program Files (x86)/Archive Utilities 2.4.0.6”

The .NET Framework 4 (full version) is required, but should already be installed on Windows 10.

After successful installation, the following facilities will be available (via desktop icons if you have chosen to create them):

- **Archive Maker**

Optionally creates, and then populates, a .CSL file with the content of a contest entry (.EDI) file.

- “Create A New Archive (.CSL file) and add .EDI file(s) to it” allows you to select a location for the Archive file (typically in the /Lists folder) and enter a suitable name for the file before saving (creating) it. A blank file is created; if this is all you want to do then click on “Cancel” instead of selecting any .EDI files.
- “Add .EDI file (or files) to an Existing Archive” Open lets you select an existing Archive to which the .EDI file(s) will be added.

In both cases, a new window opens to select the “.EDI” files. Browse to, and select, the required file(s) then click “Open”. The main programme window then shows all the contacts that are added to the file.

- **Archive Checker**

The selected archive is scanned and entries logged to the main programme window. Possible matches (where the Callsigns appear to be very similar, or Locators the same⁷⁹) are displayed.

Ticking the “Show Similar Locators” box will additionally display contacts where the Locator is similar rather than exact match.

- **Contest Reporter**

⁷⁸ There must be some limit to the number of dates retained; it is not known at present what that limit is.

⁷⁹ Note that this is often caused by one of the entries having Postcode information.

This program is intended to suggest where there may be typographic errors in your .EDI file (such as typing G9BAC instead of G9ABC). Remember that there may already be erroneous entries in the archive from earlier mis-typed QSOs.

The .EDI file for checking is selected, and then the .CSL file which will be used as a reference. The selected log is scanned and entries logged to the main programme window.

The programme distinguishes between Callsigns not found in the archive and those already present. For the latter, possible matches (where the Callsigns appear to be very similar, or Locators the same) are displayed.

Ticking the “Show Similar Locators” box will additionally display contacts where the Locator is similar rather than exact match.

Remember people do change QTH from time to time and change callsign, particularly from Foundation to Intermediate to Full etc. Good locations are also used by multiple stations, sometimes even at the same time!

Whether a problem exists is entirely up to your judgement.

- **Merge Archives**

The content of the first selected .CSL is added to the second selected CSL.

The main programme window then shows all the Entries from the first file that has been added to the second.

All of these utilities have “Close the Archive Maker Application” buttons (or “X” can be used) and (extensive) “Help”.

6.2 CSL Manager (G4APJ)

CSLmanager is a simple portable application⁸⁰ to add contacts from one or more EDI (Minos) or LOG (old G0GJV Logger) files to the designated CSL file. <<windows only at present>>

It handles (pun intended) the names of remote operators, and merges Postcode information into the archive.

“CSLmanager (V2.2.1).zip” can also be downloaded from <https://sourceforge.net/projects/minos/files/>

The Zip file includes release notes, and usage information.

⁸⁰ It may be necessary to register an OCX file if your system does not already use it; instructions are given.

7 Appendix

7.1 Suggestions

Improvements may be suggested by anyone, and if they make sense to G0GJV and, possibly after some rethinking of the request, fit in with the general philosophy for the program they may be included. Several improvements have been made at the request of users.

However, since the project is now Open Source you also make the change you want and submit it for possible inclusion in the main project.

Current possibilities include:

- There are around 25 to 30 feature requests logged
- Foreign language support for French

7.2 Future directions

- The remainder of the Applications defined in section 4.3.
- Sharing rig/rotator with other Hamlib based programs such as WSJT-X⁸¹

7.3 Known problems (as of 19 Sep 18)

This section should be considered the equivalent of 'release notes'.

- The help files are incomplete; are they sufficient, and is anyone prepared to help extend them?
- There are a few 'glitches' but please notify any problems you find.
- The RIT operation has a few inconsistencies and should probably be avoided at present.

7.4 Minos Transverter Switch

Minos Rigcontrol can be configured to send a switching message when a transverter is selected from a radio. The ASCII string message is sent out of the rigcontrol computer comport at the time of selection. A 'clear switches' message is sent when no transverter is selected.

A micro-controller such as an Arduino Uno can be utilised with a suitable relay board. The Arduino USB port is connected the computer USB.

Minos can also use a com port defined in the radio configuration tab in Minos (see Figure 70). The serial port is hard coded as 9600 Baud, 8 data, 1 stop, no parity.

Messages are sent of the form <':'><Message Body><'\n'>.

⁸¹ Currently requires full testing and documentation

<Message Body> can be any ascii sequence of characters when a transverter is selected. The values are entered in to the Switch Number box (Figure 70), available for each transverter. If this is left blank, no message will be sent.

An example interface that will probably suits most users' requirements is shown below. This software will switch relays 1 to 8. If '1' is entered in to the transverter switch number box, relay 1 will be turned on when that transverter is selected. If more than one number is entered – for example '158', then relay 1, 5 and 8 will be selected. So, it is possible to turn on more than one relay at a transverter selection, if that is required. Use '0' to clear all relays.

The 5 volt relay board is an example that is available on the internet for a reasonable price, boards with less relays can also be found if 8 aren't required. Arduino Unos are also available for a reasonable price. This document does not take you through loading the software, for which there are many tutorials on the internet to guide you through the process.

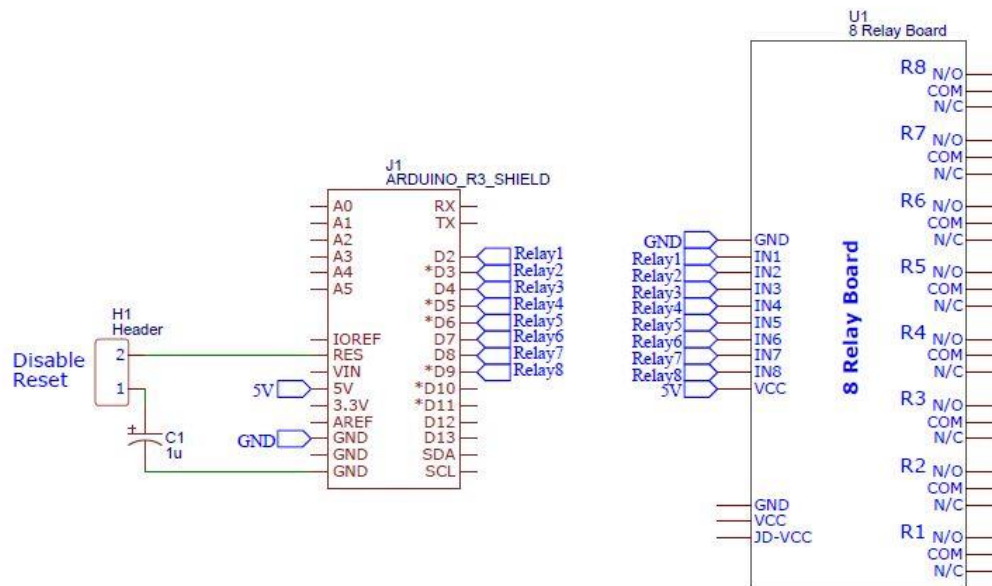


Figure 92 – Minos Transverter Switch Circuit

Figure 92 shows how to connect the IO ports to the relay board. Power for the relay board and Arduino can be taken from the computer USB port. The capacitor C1 is important, as it prevents the board performing a reset when the comport is initialised on the computer. **C1 needs to be out of circuit when uploading the switch control software to the UNO, and in circuit when the software is running.**

Inter-connections between the two boards can be achieved with the male-female Dupont jumper leads that can be purchased on the internet.

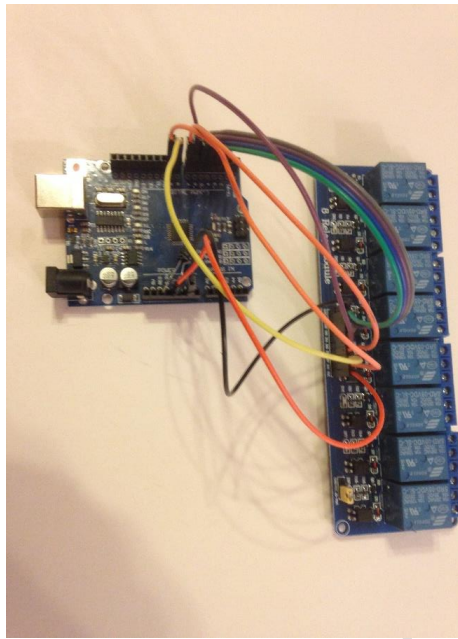


Figure 93 – Arduino and Relay Board

The Arduino Sketch can be downloaded from https://gitlab.com/dgbbal/Minos_Arduino_Transverter_Switch

7.5 Shortcut Keys

There are numerous shortcut key combinations available. They are all case-insensitive.

Rotator Application			
Function	Shortcut	Function	Shortcut
Rotate Left (CCW)	Ctrl-L	Rotate Right (CW)	Ctrl-R
Rotate to Bearing - Turn	Ctrl-T, Enter	Stop	Ctrl-S
Preset 1	Ctrl-shift-1	Preset 6	Ctrl-shift-6
Preset 2	Ctrl-shift-2	Preset 7	Ctrl-shift-7
Preset 3	Ctrl-shift-3	Preset 8	Ctrl-shift-8
Preset 4	Ctrl-shift-4	Preset 9	Ctrl-shift-9
Preset 5	Ctrl-shift-5	Preset 10	Ctrl-0

Figure 94 – Rotator Application Shortcuts

RigControl within Logger			
Function	Shortcut	Function	Shortcut
Select KHz in Frequency Box	Ctrl-f		
Select Run Frequency 1	Ctrl- shift- [	Select Run Frequency 2	Ctrl- shift-]
Memory Select 1	Ctrl-shift-1	Memory Select 6	Ctrl-shift-6

Memory Select 2	Ctrl-shift-2	Memory Select 7	Ctrl-shift-7
Memory Select 3	Ctrl-shift-3	Memory Select 8	Ctrl-shift-8
Memory Select 4	Ctrl-shift-4	Memory Select 9	Ctrl-shift-9
Memory Select 5	Ctrl-shift-5	Memory Select 10	Ctrl-shift-0
Memory Menu Read	r	Memory Menu Write	w
Memory Menu Edit	e	Memory Menu Clear	c

Figure 95 – Logger Application within Rig Control

<<Need to check all these, some no longer functional; others added?>>

[End of Document]